

Chapter 8

Spherically Symmetric Vacuum Solution

In this chapter, we examine the Schwarzschild solution—the simplest non-trivial exact solution in general relativity. This spherically symmetric solution is of great physical significance, describing both the exterior spacetime geometry of spherical astrophysical objects and serving as the fundamental model for non-rotating black holes formed through gravitational collapse.

The Einstein equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu} \quad (8.1)$$

constitute a system of highly nonlinear differential equations. For a generic matter distribution and its associated energy-momentum tensor $T_{\mu\nu}$, obtaining exact analytical solutions is exceptionally challenging. Indeed, even numerical approaches often face significant difficulties—for instance, stable and reliable simulations of spacetime dynamics during black hole collisions were not achieved until 2005.¹ Remarkably, Einstein himself initially doubted whether exact solutions could be found at all. Yet, merely months after the formulation of general relativity, Karl Schwarzschild derived the first nontrivial exact solution.² Schwarzschild’s approach was to seek a static, spherically symmetric solution. Due to the local nature of the Einstein equations, a static and spherically symmetric matter distribution

¹Numerical relativity remains an active and important subfield of general relativity. Due to scope limitations and the author’s expertise, this topic will not be covered in detail here; interested readers may consult references [53, 54].

²Karl Schwarzschild (1873–1916) was a German physicist, astronomer, and member of the Prussian Academy of Sciences. Within weeks of encountering Einstein’s equations, he obtained the spherically symmetric vacuum solution while serving as an artillery officer on the Eastern Front during World War I. His result was communicated to Einstein on December 22, 1915, and published shortly afterward on January 16, 1916. Tragically, Schwarzschild succumbed to pemphigus and died months later.

naturally implies that the exterior spacetime geometry inherits these symmetries. Thus, when studying the region outside a star, we may restrict our attention to the vacuum Einstein equations. In this chapter, we will focus exclusively on the spherically symmetric vacuum solution. Notably, it can be rigorously shown that—in four dimensions—the Schwarzschild solution is the unique spherically symmetric vacuum solution to Einstein's equations.

8.1 Stationary and Static Spacetime

In classical physics, a system is said to be in a steady state if its macroscopic properties remain stable over time, even if there is persistent motion e.g., a fluid flowing at constant velocity in a pipe. In contrast, a static system is not only time-invariant but also devoid of any internal motion or evolution—all observables are stationary in a given reference frame. However, these distinctions are reference-frame-dependent. For instance, a uniformly flowing gas appears time-independent (stationary) in a static lab frame, but in a comoving frame (moving with the fluid), the same system appears static—no motion is observed. This raises a fundamental question: How can we define stationary or static spacetimes in general relativity in a coordinate-independent, observer-invariant manner? In general relativity, the distinction is made precise through Killing vector fields.

Definition A metric is called stationary if there exists a special coordinate system in which the metric is explicitly time-independent, i.e.,

$$\left. \frac{\partial g_{\mu\nu}}{\partial x^0} \right|_* = 0, \quad (8.2)$$

where x^0 is the timelike coordinate in this coordinate system. This essentially means that ∂_{x^0} is a Killing vector. In short, a spacetime is called stationary if and only if it has a timelike Killing vector field.

For a stationary spacetime, by choosing a suitable coordinate system, the metric can always be expressed in the form:

$$ds^2 = g_{00}(dx^0)^2 + g_{0i}dx^0dx^i + g_{ij}dx^idx^j, \quad (8.3)$$

where all the metric components are independent of the timelike coordinate. Note that $g_{0i} \neq 0$ is allowed, indicating that the spacetime may still possess evolution. If we further impose the condition that the spacetime is static,

meaning the metric remains invariant under the transformation $x^0 \rightarrow x'^0 = -x^0$, we must have

$$g_{\mu\nu} = g'_{\mu\nu} \Rightarrow g_{0i} = 0. \quad (8.4)$$

Thus, a general form of the metric for a static spacetime can be written as

$$ds^2 = g_{00}(dx^0)^2 + g_{ij}dx^i dx^j. \quad (8.5)$$

To provide a coordinate-independent characterization of a static spacetime, we introduce the concept of a hypersurface orthogonal vector. A hypersurface can be defined by a family of equations $f(x^\mu) = c$. At a point P on this hypersurface, the components of the normal covector are given by $n_\mu \equiv \frac{\partial f}{\partial x^\mu}$. A vector field $\hat{X}$ is said to be hypersurface orthogonal if it is everywhere orthogonal to such a family of hypersurfaces, i.e.,

$$X_\mu = \lambda(x)n_\mu = \lambda f_{,\mu},$$

which leads to

$$X_\mu \partial_\nu X_\sigma = \lambda f_{,\mu} \partial_\nu (\lambda f_{,\sigma}) = \lambda f_{,\mu} \lambda_{,\nu} f_{,\sigma} + \lambda^2 f_{,\mu} f_{,\nu\sigma}. \quad (8.6)$$

Antisymmetrization this expression yields

$$X_{[\mu} \partial_\nu X_{\sigma]} = 0 \Rightarrow X_{[\mu} \nabla_\nu X_{\sigma]} = 0. \quad (8.7)$$

Any hypersurface orthogonal vector field satisfies Eq. (8.7). Conversely, any non-zero Killing vector field satisfying Eq. (8.7) must be hypersurface orthogonal.

Theorem A spacetime is static if and only if it has a hypersurface orthogonal timelike Killing vector field.

Proof For a stationary spacetime, there exists a timelike Killing vector field $X^\mu = \delta_0^\mu$:

$$X_\mu = g_{\mu\nu} X^\nu = g_{\mu 0}, \quad |X|^2 = X_\mu X^\mu = g_{\mu 0} \delta_0^\mu = g_{00}. \quad (8.8)$$

If further, this Killing vector field is hypersurface orthogonal,

$$X_\mu = |X|^2 f_{,\mu}, \quad (8.9)$$

where $f_{,\mu}$ is required to be normalized, then we have

$$g_{0\mu} = g_{00} f_{,\mu}. \quad (8.10)$$

For $\mu = 0$, $f_{,0} = 1$, so

$$f = x^0 + h(x^i). \quad (8.11)$$

Consider the coordinate transformation

$$x^0 \rightarrow x'^0 = x^0 + h(x^i), \quad x^i \rightarrow x'^i = x^i, \quad (8.12)$$

a vector field transforms as

$$X'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} X^\nu = \delta_0^\mu, \quad (8.13)$$

and the metric tensor field transforms as

$$g'_{\mu\nu,0} = 0, \quad g'_{00} = g_{00}, \quad (8.14)$$

while

$$\begin{aligned} g'_{0i} &= \frac{\partial x^\mu}{\partial x'^0} \frac{\partial x^\nu}{\partial x'^i} g_{\mu\nu} \\ &= \frac{\partial x^0}{\partial x'^0} \left(\frac{\partial x^0}{\partial x'^i} g_{00} + \frac{\partial x^j}{\partial x'^i} g_{0j} \right) \\ &= -\partial_i h g_{00} + g_{0i}. \end{aligned} \quad (8.15)$$

We can choose $\partial_i h = g_{0i}/g_{00}$ to make $g'_{0i} = 0$. In this new coordinate system, $g_{0i} = 0$, and the metric is static.

In the preceding analysis, the static metric retains a residual coordinate freedom of the form:

$$x^0 \rightarrow x'^0 = Ax^0 + B, \quad x^i \rightarrow x'^i = h'^i(x^j), \quad (8.16)$$

where A and B are constants and h'^i is an arbitrary spatial coordinate transformation. These transformations do not change the above discussion, and the obtained metric still maintains the form of a static metric. If the boundary condition requires that $g_{00} \rightarrow 1$ at spatial infinity, then $A = 1$. In this way, we define a time coordinate called worldline time. Therefore, for a static spacetime, we regain the idea of absolute time in Newtonian gravity, that is, the spacetime can be well sliced by hypersurfaces $t = \text{constant}$.

8.2 Birkhoff's Theorem

For the vacuum Einstein equations, the spherically symmetric solution is unique. More precisely, there is the following theorem.

Birkhoff theorem The four-dimensional Schwarzschild solution is the unique spherically symmetric solution to the vacuum Einstein equations.

Below, we outline a proof of this theorem. The term “spherical symmetry” in Birkhoff’s theorem means that the solution possesses the same symmetry as a two-dimensional sphere. Specifically, a two-dimensional sphere

S^2 admits three Killing vector fields ($V^{(1)}$, $V^{(2)}$, $V^{(3)}$), which satisfy the $so(3)$ Lie algebra:

$$\begin{aligned} [V^{(1)}, V^{(2)}] &= V^{(3)}, \\ [V^{(2)}, V^{(3)}] &= V^{(1)}, \\ [V^{(3)}, V^{(1)}] &= V^{(2)}. \end{aligned} \quad (8.17)$$

To proceed, we invoke Frobenius' theorem in differential geometry: Given a set of mutually commuting vector fields, then there exists a coordinate system in which these vector fields correspond to the partial derivatives along the coordinate directions. In other words, these vector fields generate translations in these coordinates, effectively defining a Cartesian-like coordinate system. More generally, if these vector fields do not commute but form a closed algebra, their integral curves collectively describe a submanifold. On this submanifold, the vector fields are well-defined and generate its isometry group. The submanifold's dimension is at most equal to the number of vector fields, as the latter cannot span a space of higher dimension. Crucially, since the vector fields are globally defined on the manifold, every point lies on some such submanifold. In the context of spherically symmetric space-time, the two-dimensional sphere arises as the submanifold generated by the Killing vector fields. The full spacetime may be viewed as a foliation, with the spherical symmetry “leafing” out from this two-dimensional base.

Example 8.1. Three-dimensional Euclidean space $\mathbf{R}^3$.

Three-dimensional Euclidean space $\mathbf{R}^3$ is obviously spherically symmetric, which can be clearly seen using spherical coordinates. In spherical

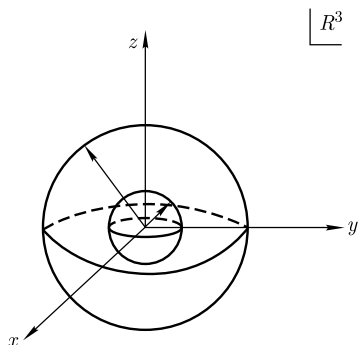


Fig. 8.1 Using spherical coordinates to foliate three-dimensional Euclidean space: at different radial positions, there is a two-dimensional sphere.

coordinates, for each fixed radial coordinate, we have a two-dimensional sphere, as shown in Figure 8.1.

Example 8.2. A wormhole with topology $\mathbf{R} \times \mathbf{S}^2$.

In this case, the spacetime exhibits spherical symmetry without a fixed origin about which the geometry rotates. Instead, the entire manifold is constructed by “attaching” the remaining dimensions to the two-dimensional sphere, as shown in Figure 8.2.

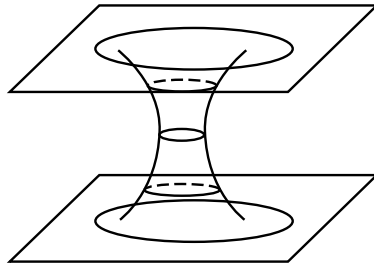


Fig. 8.2 A schematic diagram of a wormhole embedded in a higher dimension. For display convenience, each circle actually represents a two-dimensional sphere. The spacetime can be foliated along the vertical direction.

In general, consider an n -dimensional manifold foliated on an m -dimensional submanifold. We can describe such a structure using coordinate functions u^i on the m -dimensional submanifold and $n - m$ additional coordinates v^I to parametrize the transverse directions, thereby specifying the position within the foliation. The coordinate functions u^i and v^I together describe the full manifold. When the submanifold is a maximally symmetric space, such as the spherically symmetric case under consideration, a powerful simplification arises: there exists a choice of u coordinates such that the metric of the entire manifold takes the following block-diagonal form:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g_{IJ}(v) dv^I dv^J + f(v) \gamma_{ij}(u) du^i du^j, \quad (8.18)$$

where $\gamma_{ij}(u)$ is the metric on the maximally symmetric submanifold. Note that there are no cross terms like $dv^I du^j$ in the above metric, and $g_{IJ}(v)$ and $f(v)$ are only functions of v^I , independent of u^i .

For a spherically symmetric spacetime, using coordinate transformations, we can always start from the general metric ansatz

$$ds^2 = m(t, r) dt^2 + n(t, r) dr^2 + r^2 d\Omega^2 \quad (8.19)$$

to solve the vacuum Einstein equations. Here $m(t, r)$ and $n(t, r)$ are arbitrary functions, and $d\Omega^2$ is the metric on the unit 2-sphere. At this stage, the coordinates (t, r) are merely labels with no predetermined geometric interpretation. To fix the physical meaning of these coordinates, we compare (8.19) with the Minkowski spacetime metric in spherical coordinates:

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2. \quad (8.20)$$

Far from any gravitational source (e.g., a star), we expect spacetime to approach flatness. This asymptotic flatness motivates identifying r as the radial coordinate and t as the time coordinate, with $m(t, r)$ becoming negative to preserve the timelike nature of t . For practical calculations, it is often more convenient to adopt an exponential parametrization of the metric components:

$$ds^2 = -e^{2\alpha(t,r)} dt^2 + e^{2\beta(t,r)} dr^2 + r^2 d\Omega^2. \quad (8.21)$$

Note that this metric assumption comes entirely from the requirement of spherical symmetry and is independent of the Einstein equations, so it also applies to solving the spherically symmetric solutions of the non-vacuum Einstein equations. The non-zero Christoffel symbols corresponding to this metric are

$$\begin{aligned} \Gamma_{00}^0 &= \partial_0 \alpha, & \Gamma_{01}^0 &= \partial_1 \alpha, & \Gamma_{11}^0 &= e^{2(\beta-\alpha)} \partial_0 \beta, \\ \Gamma_{00}^1 &= e^{2(\alpha-\beta)} \partial_1 \alpha, & \Gamma_{01}^1 &= \partial_0 \beta, & \Gamma_{11}^1 &= \partial_1 \beta, \\ \Gamma_{12}^2 &= \frac{1}{r}, & \Gamma_{22}^1 &= -re^{-2\beta}, & \Gamma_{13}^3 &= \frac{1}{r}, \\ \Gamma_{33}^1 &= -re^{-2\beta} \sin^2 \theta, & \Gamma_{33}^2 &= -\sin \theta \cos \theta, & \Gamma_{23}^3 &= \frac{\cos \theta}{\sin \theta}, \end{aligned}$$

where $(0, 1, 2, 3)$ represent (t, r, θ, ϕ) respectively. From these, we can obtain the components of the Riemann tensor:

$$\begin{aligned} R_{101}^0 &= e^{2(\beta-\alpha)} [\partial_0^2 \beta + (\partial_0 \beta)^2 - \partial_0 \alpha \partial_0 \beta] \\ &\quad + [\partial_1 \alpha \partial_1 \beta - \partial_1^2 \alpha - (\partial_1 \alpha)^2], \\ R_{202}^0 &= -re^{-2\beta} \partial_1 \alpha, \\ R_{303}^0 &= -re^{-2\beta} \sin^2 \theta \partial_1 \alpha, \\ R_{212}^0 &= -re^{-2\alpha} \partial_0 \beta, \\ R_{313}^0 &= -re^{-2\alpha} \sin^2 \theta \partial_0 \beta, \\ R_{212}^1 &= re^{-2\beta} \partial_1 \beta, \\ R_{313}^1 &= re^{-2\beta} \sin^2 \theta \partial_1 \beta, \\ R_{323}^2 &= (1 - e^{-2\beta}) \sin^2 \theta, \end{aligned} \quad (8.22)$$

and the components of the Ricci tensor:

$$\begin{aligned}
R_{00} &= [\partial_0^2 \beta + (\partial_0 \beta)^2 - \partial_0 \alpha \partial_0 \beta] + e^{2(\alpha-\beta)} \\
&\quad \left[\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta + \frac{2}{r} \partial_1 \alpha \right], \\
R_{11} &= - \left[\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta - \frac{2}{r} \partial_1 \beta \right] \\
&\quad + e^{2(\beta-\alpha)} [\partial_0^2 \beta + (\partial_0 \beta)^2 - \partial_0 \alpha \partial_0 \beta], \\
R_{01} &= \frac{2}{r} \partial_0 \beta, \\
R_{22} &= e^{-2\beta} [r(\partial_1 \beta - \partial_1 \alpha) - 1] + 1, \\
R_{33} &= R_{22} \sin^2 \theta.
\end{aligned} \tag{8.23}$$

We only need to solve the vacuum Einstein equations $R_{\mu\nu} = 0$. From $R_{01} = 0$, we find

$$\partial_0 \beta = 0 \quad \Rightarrow \quad \beta = \beta(r), \tag{8.24}$$

that is, β must be a function of r . Therefore, R_{00} and R_{11} are further simplified, leading to the following two equations:

$$\begin{aligned}
e^{2(\alpha-\beta)} \left[\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta + \frac{2}{r} \partial_1 \alpha \right] &= 0, \\
\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta - \frac{2}{r} \partial_1 \beta &= 0.
\end{aligned} \tag{8.25}$$

This leads to

$$\partial_1(\alpha + \beta) = 0 \quad \Rightarrow \quad \alpha + \beta = g(t), \tag{8.26}$$

where $g(t)$ is an arbitrary function of time. That is to say,

$$\alpha = -\beta(r) + g(t). \tag{8.27}$$

The metric can therefore be expressed in the form:

$$ds^2 = -e^{-2\beta(r)+2g(t)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2. \tag{8.28}$$

We still have the freedom to redefine the time coordinate $dt \rightarrow e^{-g(t)} dt$ such that $g(t) = 0$. This simplifies the metric to

$$ds^2 = -e^{-2\beta(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2, \tag{8.29}$$

where all the metric components are now explicitly t -independent. In fact, we have already proven an important fact: any spherically symmetric vacuum solution necessarily admits a timelike Killing vector field ∂_t , and this metric is not only stationary but also static.

Furthermore, from $R_{22} = 0$, we obtain

$$\begin{aligned} e^{2\alpha}(2r\partial_1\alpha + 1) &= 1 \\ \Rightarrow \partial_1(re^{2\alpha}) &= 1 \\ \Rightarrow e^{2\alpha} &= 1 + \frac{C}{r}, \end{aligned} \quad (8.30)$$

where C is an undetermined integration constant. Finally, we obtain the metric for a spherically symmetric spacetime in vacuum

$$ds^2 = -\left(1 + \frac{C}{r}\right) dt^2 + \left(1 + \frac{C}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (8.31)$$

As a consistency check, we must verify whether this metric satisfies $R_{00} = 0$. Indeed, it can be confirmed that this condition holds.

As previously discussed, the spherically symmetric vacuum solution describes the spacetime geometry outside a spherically symmetric star. In the limit $r \rightarrow \infty$, where the gravitational influence of the star diminishes, we expect the metric to reduce to the Newtonian limit. Specifically, the metric components should satisfy:

$$\begin{aligned} g_{00} &= -\left(1 + \frac{C}{r}\right) \sim g_{00}|_{\text{Newton}} = -(1 + 2\Phi), \\ g_{rr} &= \left(1 + \frac{C}{r}\right)^{-1} \approx 1 - \frac{C}{r} \sim g_{rr}|_{\text{Newton}} = 1 - 2\Phi, \end{aligned} \quad (8.32)$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential. By comparing these expressions, we identify the integration constant C as $C = -2GM$. Substituting this back into the metric, we arrive at the well-known Schwarzschild metric:

Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (8.33)$$

When $M \rightarrow 0$, corresponding to the absence of a central mass, the Schwarzschild metric reduces to the Minkowski spacetime, as expected for flat spacetime in special relativity. The form of the metric presented earlier assumes natural units, $c = 1$. Restoring dimensional consistency, the Schwarzschild metric including the speed of light c is given by

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (8.34)$$

In many literatures, some authors introduce the shorthand $\mu = GM/c^2$, which represents the geometric mass (with units of length). It is also common to adopt units where $G = 1$ (and often $c = 1$), simplifying the metric further while retaining physical clarity.

From the form of the Schwarzschild metric, it is evident that the metric appears singular at $r = 0$ and $r = 2GM$. However, the explicit expression of the metric is highly dependent on the choice of coordinates, or more precisely, on the observer's reference frame. These apparent singularities may not represent true physical singularities but could instead be artifacts of the coordinate system, known as coordinate singularities.

For example, consider the polar coordinate system in two-dimensional Euclidean space, where the metric is given by $ds^2 = dr^2 + r^2 d\theta^2$. At the origin $r = 0$, the metric becomes degenerate, and the component $g^{\theta\theta} = r^{-2}$ diverges. Nevertheless, the underlying geometry of the plane is everywhere flat, and no genuine singularity exists. The apparent singularity in this case arises solely from the choice of coordinates. Similarly, in the Schwarzschild metric, the singularity at $r = 2GM$ is merely a coordinate singularity, as can be verified by transforming to alternative coordinate systems, where the metric remains well-behaved at this radius. The details of such coordinate transformations will be discussed in later chapter.

To determine whether a singularity is genuine (i.e., a physical singularity) rather than a mere artifact of the coordinate system, we can employ the following criterion: a true singularity occurs where curvature invariants diverge. The curvature invariants are scalar quantities constructed from the Riemann tensor, Ricci tensor, Ricci scalar, covariant derivatives, and their contractions, such as

$$R, \quad R_2 = R^{\mu\nu} R_{\mu\nu}, \quad R_4 = R^{\mu\nu\sigma\rho} R_{\mu\nu\sigma\rho}, \quad R_{\mu\nu\rho\sigma} R^{\rho\sigma\lambda\tau} R^{\mu\nu}_{\lambda\tau}, \dots$$

If any of these scalars diverges at a certain place, that place constitutes a true spacetime singularity. However, this is a sufficient but not necessary condition, as some singularities (e.g., conical singularities) may not exhibit divergent curvature scalars. Another method to identify singularities is by examining geodesic completeness—whether freely falling particles or light rays can be extended indefinitely in proper time or affine parameter. In most cases, the divergence of curvature invariants and geodesic incompleteness coincide in signaling a true singularity. For the Schwarzschild metric, since $R_{\mu\nu} = 0$, $R = 0$, $R_2 = 0$, however

$$R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = \frac{48G^2 M^2}{r^6} \quad (8.35)$$

diverges as $r \rightarrow 0$. This confirms that $r = 0$ is a true curvature singularity. In contrast, at $r = 2GM$, no curvature invariants diverge, indicating that this is merely a coordinate singularity. Nevertheless, this radius is physically significant: it defines the event horizon of a Schwarzschild black hole, known as the Schwarzschild horizon or Schwarzschild radius:

$$R_s = \frac{2GM}{c^2}. \quad (8.36)$$

From the above analysis, it is evident that our choice of the constant $C = -2GM$ is motivated by matching the solution to the Newtonian approximation in the weak-field limit. From a purely mathematical standpoint, the constant C remains arbitrary. In particular, selecting $C = 2GM$ with $M > 0$ leads to a spacetime metric that is regular everywhere except at the central singularity $r = 0$. In this case, the coordinate singularity associated with the Schwarzschild horizon vanishes, resulting in a crucial physical implication: while $r = 0$ is still a curvature singularity, the absence of an event horizon exposes this singularity, giving rise to a naked singularity. As we shall discuss later, the existence of a naked singularity is unphysical.

The Schwarzschild metric has several important properties:

(1) Asymptotically flatness.

The Schwarzschild metric is asymptotically flat: as $r \rightarrow \infty$, it converges to the Minkowski metric. This reflects the physical expectation that far from the gravitating source (e.g., a star or black hole), spacetime curvature diminishes, and the geometry approaches that of flat spacetime.

(2) Schwarzschild coordinates.

While the Schwarzschild spacetime, as a solution to the vacuum Einstein equations, is coordinate-independent, the metric (8.34) employs a specific choice of coordinates (t, r, θ, ϕ) , known as Schwarzschild coordinates. These coordinates are physically meaningful: at radial infinity, they align with the frame of a static observer, consistent with the asymptotic Minkowski limit.

(3) Stationarity and staticity.

Outside a star or a black hole's event horizon, the Schwarzschild spacetime is static, as the metric components are time-independent and ∂_t is a time-like Killing vector. However, for $r < R_s$, ∂_t becomes spacelike, although it remains a Killing vector of spacetime. This distinction has profound implications for the causal structure of black holes, which we will explore later when discussing the Schwarzschild black hole.

We have actually proven Birkhoff's theorem: the Schwarzschild metric is the unique spherically symmetric solution to the vacuum Einstein

equations. Notably, our derivation imposes no prior assumptions about stationarity or staticity, nor does it depend on the nature of the source generating the spacetime. Even if the central object (e.g., a star) undergoes time-dependent spherical evolution—such as gravitational collapse—the external spacetime remains static and described by the Schwarzschild metric. In other words, no time-varying, spherically symmetric vacuum solutions exist; all such solutions must reduce to Schwarzschild.

The Schwarzschild metric exhibits remarkable uniqueness properties in general relativity, as highlighted by the following theorems.

Israel’s theorem Any static vacuum solution to Einstein’s field equations must be spherically symmetric and is therefore described by the Schwarzschild metric.

Thus, we have the following uniqueness theorem.³

Uniqueness theorem The Schwarzschild geometry is the unique static vacuum solution in general relativity.

While the Schwarzschild spacetime may appear in different forms due to coordinate choices, it is always reducible to the standard Schwarzschild metric via a suitable coordinate transformation. Historically, this has led to its “rediscovery” over twenty times in various coordinate systems.

8.3 Geodesics in Schwarzschild Spacetime

In this section, we examine the geodesic motion of particles in Schwarzschild spacetime. Our analysis encompasses both massive and massless particles, whose distinct kinematic behaviors yield different observational signatures, including those relevant to the classical tests of general relativity. Furthermore, investigating particle geodesics provides crucial insights into the fundamental properties of Schwarzschild spacetime, particularly those characterizing Schwarzschild black holes.

The geodesic equation governing the motion of a particle in spacetime

³A similar uniqueness theorem holds in higher dimensions under the assumptions that the theory is still Einstein’s general theory of relativity and that the action has no higher derivative term. For a detailed discussion, see S. Hollands and A. Ishibashi, *Black Hole Uniqueness Theorems in Higher Dimensional Spacetime*. *Class. Quant. Grav.*, 2012, 29: 163001.

is given by

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0,$$

where λ is an affine parameter along the trajectory. Using the Christoffel symbols for the Schwarzschild metric

$$\begin{aligned} \Gamma_{00}^1 &= \frac{GM}{r^3}(r - R_s), & \Gamma_{11}^1 &= \frac{-GM}{r(r - R_s)}, & \Gamma_{01}^0 &= \frac{GM}{r(r - R_s)}, \\ \Gamma_{12}^2 &= \frac{1}{r}, & \Gamma_{22}^1 &= -(r - R_s), & \Gamma_{13}^3 &= \frac{1}{r}, \\ \Gamma_{33}^1 &= -(r - R_s) \sin^2 \theta, & \Gamma_{33}^2 &= -\sin \theta \cos \theta, & \Gamma_{23}^3 &= \frac{\cos \theta}{\sin \theta}, \end{aligned} \quad (8.37)$$

we obtain a set of geodesic equations:

$$\frac{d^2 t}{d\lambda^2} + \frac{R_s}{r(r - R_s)} \frac{dr}{d\lambda} \frac{dt}{d\lambda} = 0, \quad (8.38)$$

$$\begin{aligned} \frac{d^2 r}{d\lambda^2} + \frac{GM}{r^3} r(r - R_s) \left(\frac{dt}{d\lambda} \right)^2 - \frac{GM}{r(r - R_s)} \left(\frac{dr}{d\lambda} \right)^2 \\ - (r - R_s) \left[\left(\frac{d\theta}{d\lambda} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right] = 0, \end{aligned} \quad (8.39)$$

$$\frac{d^2 \theta}{d\lambda^2} + \frac{2}{r} \frac{d\theta}{d\lambda} \frac{dr}{d\lambda} - \sin \theta \cos \theta \left(\frac{d\phi}{d\lambda} \right)^2 = 0, \quad (8.40)$$

$$\frac{d^2 \phi}{d\lambda^2} + \frac{2}{r} \frac{d\phi}{d\lambda} \frac{dr}{d\lambda} + 2 \frac{\cos \theta}{\sin \theta} \frac{d\theta}{d\lambda} \frac{d\phi}{d\lambda} = 0. \quad (8.41)$$

At first glance, these coupled second-order differential equations appear highly intricate and analytically challenging. However, the spherical symmetry of the Schwarzschild spacetime allows for significant simplifications. Rather than solving them directly, we exploit the conserved quantities associated with the spacetime symmetries, namely, energy and angular momentum, to reduce the problem to a more tractable form.

The full isometry group of Schwarzschild spacetime is simply $U(1) \times SO(3)$, where the $U(1)$ factor arises from the time-translation symmetry and $SO(3)$ from spherical symmetry. The timelike Killing vector is $\hat{\xi} = \partial_t$ (for $r > R_s$), with components $\xi^\mu = (1, 0, 0, 0)$. Among the three spacelike Killing vectors, the most apparent one is $\hat{\eta} = \partial_\phi$, representing azimuthal symmetry. Using the Killing vectors, we can define conserved quantities for

a particle in geodesic motion: the inner product of the Killing vector field $\hat{K}$ and the tangent vector $\hat{u}$ remains constant along the geodesic:

$$K_\mu \frac{dx^\mu}{d\lambda} = \text{constant}, \quad (8.42)$$

where λ is the proper time for massive particles and the affine parameter for massless particles.

First, using the timelike Killing vector $\xi^\mu = (1, 0, 0, 0)$, we can define the conserved energy density of the particle. From the expressions $\xi_\mu = \left(-\left(1 - \frac{R_s}{r}\right), 0, 0, 0\right)$, $u^\mu = \frac{dx^\mu}{d\lambda}$, we derive the conserved quantity

$$E = \left(1 - \frac{R_s}{r}\right) \frac{dt}{d\lambda}. \quad (8.43)$$

Indeed, this satisfies $u_0 = g_{0\mu}u^\mu = -E$. For a massive particle following a timelike geodesic, as $r \rightarrow \infty$, we have $\|\hat{\xi}\|^2 \sim -1$, and E corresponds to the total energy per unit mass as measured by a static observer at infinity. This is because such an observer would measure the particle's energy as $E_o = -\hat{u}_{\text{obs}} \cdot \hat{P} = mE$. Generally speaking, relative to a static observer at infinity, E represents the total energy per unit mass of a particle moving along the geodesic, incorporating both kinetic and gravitational potential energy contributions. Physically, this can be viewed as the energy required by the observer to place a unit mass particle into a specific orbital configuration, effectively, the particle's energy density. Notably, if the particle is released from rest at infinity, $E = 1$. In the case of a massless particle (e.g., a photon) following a null geodesic, E corresponds to its “total energy”. However, it is important to note that due to the freedom in choosing the affine parameter for null geodesics, this “total energy” lacks a precise physical meaning—it is inherently arbitrary and does not represent a measurable physical quantity.

Secondly, the $SO(3)$ spatial rotation symmetry allows us to define the conserved angular momentum of the particle. Consider the spacelike Killing vector field $\hat{\eta} = \partial_\phi$, or equivalently $\eta_\mu = (0, 0, 0, r^2 \sin^2 \theta)$. The corresponding conserved quantity for the particle is

$$L = g^{\mu\nu} \eta_\mu u_\nu = r^2 \sin^2 \theta \frac{d\phi}{d\lambda}. \quad (8.44)$$

For a massive particle, L represents the angular momentum per unit mass (angular momentum density) as measured by a static observer at infinity. For a massless particle, L corresponds to its “orbital angular momentum”. In addition to ∂_ϕ , spherically symmetric spacetime possesses two other Killing vectors associated with rotational symmetry. The conserved quantities they define collectively correspond to the three-dimensional angular

momentum vector. The invariance of this vector's direction implies that the particle's motion is confined to a fixed orbital plane. Without loss of generality, we can choose this plane to be the equatorial plane $\theta = \frac{\pi}{2}$. This can also be deduced from the geodesic equation for θ :

$$\frac{d}{d\lambda} \left(r^2 \frac{d\theta}{d\lambda} \right) - \frac{L^2 \cos \theta}{r^2 \sin^3 \theta} = 0. \quad (8.45)$$

Integrating this equation yields

$$\left(r^2 \frac{d\theta}{d\lambda} \right)^2 = -L^2 \cot^2 \theta + C, \quad (8.46)$$

where C is an integration constant. Using spherical symmetry, we can always align the initial conditions such that at some $\lambda = \lambda_0$, $\theta(\lambda_0) = \frac{\pi}{2}$, $\frac{d\theta}{d\lambda}(\lambda_0) = 0$, meaning the particle starts in the equatorial plane with no initial polar velocity. This fixes $C = 0$. Consequently, (8.46) holds only if both sides vanish identically, leading to

$$\theta(\lambda) = \frac{\pi}{2}, \quad \frac{d\theta}{d\lambda}(\lambda) = 0. \quad (8.47)$$

Thus, the particle's trajectory remains confined to the equatorial plane for all time. In what follows, we restrict our analysis to $\theta = \frac{\pi}{2}$, where the angular momentum density reduces to

$$L = r^2 \frac{d\theta}{d\lambda}. \quad (8.48)$$

Furthermore, we can utilize the normalization condition of the particle's 4-velocity to simplify our analysis. The compatibility of the metric ensures that the inner product of the particle's 4-velocity along the geodesic remains constant,

$$\epsilon = -g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}, \quad (8.49)$$

where, for massive particles (parametrized by proper time) $\epsilon = 1$, and for massless particles $\epsilon = 0$. For the Schwarzschild metric, this condition gives

$$-\left(1 - \frac{R_s}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{R_s}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon.$$

Substituting the conserved quantities E and L , we obtain the radial equation of motion:

$$-E^2 + \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{R_s}{r}\right) \left(\frac{L^2}{r^2} + \epsilon\right) = 0. \quad (8.50)$$

This can be reformulated into a more physically intuitive form, resembling the energy equation for classical motion:

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + V(r) = \frac{1}{2} E^2, \quad (8.51)$$

where the effective potential $V(r)$ is given by

$$V(r) = \frac{1}{2} \epsilon - \epsilon \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (8.52)$$

Equation (8.51) is analogous to the classical description of a unit-mass particle with “total energy” of $\frac{1}{2} E^2$, moving in a one-dimensional potential with potential energy $V(r)$. The effective potential energy $V(r)$ includes contributions from the particle’s rest mass (ϵ -dependent term), gravitational attraction ($-GM/r$), centrifugal barrier ($L^2/2r^2$), and general relativistic correction ($-GML^2/r^3$). This equation fully determines the radial motion $r(\lambda)$ of the particle. The angular motion $\phi(\lambda)$ and coordinate time evolution $t(\lambda)$ can then be obtained straightforwardly using the conserved quantities E and L .

The effective potential not only depends on the properties of spacetime itself but also on the particle’s mass and conserved angular momentum density. In the effective potential $V(r)$, the second term corresponds to the usual Newtonian gravitational potential, the third term represents the centrifugal potential, and the last term is the correction term from general relativity. The correction term becomes significant at small r . Without it, the equation reduces to the radial motion of a point particle in Newtonian gravity. The qualitative behavior of the particle’s radial motion can be understood by comparing $\frac{1}{2} E^2$ with $V(r)$. Consider a particle initially moving inward from infinity. The possible scenarios are as follows:

(1) Bound motion.

Here $\frac{1}{2} E^2$ intersects $V(r)$ at two points, confining the particle’s motion between them. In Newtonian mechanics, this corresponds to an elliptical orbit. If the two intersection points merge, meaning $\frac{1}{2} E^2$ coincides with a local minimum of $V(r)$, the particle undergoes stable circular motion.

(2) Unbound motion with a turning point.

If $\frac{1}{2} E^2$ intersects $V(r)$ only once, the particle reaches a turning point and reverses direction, escaping back to infinity.

(3) Captured by the star.

When $\frac{1}{2} E^2$ does not intersect $V(r)$, i.e., the particle’s total energy density

exceeds the potential barrier's peak, the particle continues inward until it is captured by the star. Notably, if $\frac{1}{2}E^2$ precisely matches a local maximum of $V(r)$, the particle may briefly execute unstable circular motion before either falling inward or escaping.

In the following discussion, we will treat the central celestial body as a point mass, effectively modeling the Schwarzschild spacetime as a black hole by neglecting its finite radius.

Let us first look at motions of particles in Newtonian gravity. The effective potential for a massive particle in Newtonian gravity is given by:

$$V(r) = \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2}. \quad (8.53)$$

This potential admits circular orbits at the radius

$$r_c = \frac{L^2}{GM}, \quad (8.54)$$

with a corresponding binding energy of $E = -\frac{GM}{2r_c}$. A particle starting from rest at infinity must lose energy to settle into such an orbit. Beyond circular motion, elliptical orbits (bound trajectories) are also possible, depending on the particle's energy. For massless particles, the potential reduces to

$$V(r) = \frac{L^2}{2r^2}. \quad (8.55)$$

Unlike massive particles, photons cannot form circular or bound orbits in Newtonian gravity. A photon arriving from infinity will gradually "slow down" before being reflected back to infinity. In Newtonian gravity, massless particles are not affected by gravitational attraction, and their trajectories are straight lines.

In general relativity, the trajectories of particles deviate significantly from those predicted by Newtonian mechanics. For massive particles, the effective potential governing their motion is given by

$$V(r) = \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (8.56)$$

Notably, as $r \rightarrow 0$, the potential $V \rightarrow -\infty$, a behavior starkly contrasting with Newtonian gravity. At $r = R_s$, the potential vanishes $V = 0$. The extremal points of the potential, corresponding to circular orbits, are determined by solving

$$\begin{aligned} 0 &= \frac{\partial V}{\partial r} = r^{-4}[GM r^2 - L^2 r + 3GML^2] \\ \Rightarrow R_{\pm} &= \frac{L^2 \pm \sqrt{L^4 - 12L^2 M^2 G^2}}{R_s}. \end{aligned} \quad (8.57)$$

If the angular momentum is small, $L^2 < 12G^2M^2$, no extremal points exist. The inward-moving particle $dr/d\lambda < 0$ will inevitably plunge toward the central object, either colliding with a compact star or crossing the event horizon of a black hole and heading towards the center of the black hole. If the particle's angular momentum is large enough, $L^2 > 12G^2M^2$, the potential admits two critical radii: a minimum at R_+ supporting stable circular orbits; a maximum at R_- , permitting unstable circular orbits. In the limit $L \gg GM$, the stable orbit radius reduces to $R_+ \sim L^2/GM$, recovering the Newtonian result. Note that as $R_+ > 6GM$, there are no stable circular orbits smaller than $6GM$, while unstable circular orbits are confined to $3GM < R_- < 6GM$. For a given circular orbit R_c , the required angular momentum is

$$L = \left(\frac{GMR_c}{1 - 3GM/R_c} \right)^{\frac{1}{2}}. \quad (8.58)$$

For the circular orbit $dr/d\lambda = 0$, the conserved energy density of the particle satisfies

$$\frac{1}{2}E^2(R) = V(R) \Rightarrow E(R) = \frac{R - R_s}{\sqrt{R(R - 3GM)}}. \quad (8.59)$$

When $R \leq 4GM$, $E \geq 1$, particles may escape to infinity if perturbed, though they risk capture by the central object. As $R \rightarrow 3GM$, then $E, L \rightarrow \infty$. If $4GM < R < 6GM$, since $E < 1$, particle remain confined to bound trajectories.

When $L^2 = 12G^2M^2$, a special case arises: the stable and unstable circular orbits coincide at a single radius, $R_+ = R_- = 6GM$. This unique orbit is known as the innermost stable circular orbit (ISCO) and plays a crucial role in astrophysics, particularly in the study of accretion disks around compact objects. In the vicinity of a massive compact star or black hole, an accretion disk forms as gas gradually spirals inward due to turbulent viscosity—often driven by magnetohydrodynamic instabilities. As the gas loses angular momentum, it migrates inward, converting gravitational potential energy into heat. Eventually, once the gas reaches the critical angular momentum $L < L_c$, it can no longer perform circular motion and rapidly plunges toward the central object. The ISCO marks the final stable boundary where orbiting matter can resist this inward pull. Beyond this point, no further stable orbits exist, and the gas is inevitably captured. At the ISCO, the particle's energy density is $E^2 = 8/9$. For a particle originating from infinity where $E = 1$, the binding energy per unit mass at the ISCO is

$$E_B = 1 - E = 1 - \sqrt{8/9} \approx 0.0572. \quad (8.60)$$

This corresponds to 5.7% of the particle's rest mass $m_0 c^2$ being released before final capture. Such an energy loss is extraordinarily efficient compared to other astrophysical processes. For instance, nuclear fusion efficiency (hydrogen to helium), the latter's efficiency is only about 0.7%. The majority of this energy is expected to be emitted as gravitational waves, particularly near the ISCO, where the orbital dynamics are most extreme.

For massless particles, $\epsilon = 0$, the predictions of general relativity differ markedly from Newtonian gravity. The effective potential energy governing their motion takes the form

$$V(r) = \frac{L^2}{2r^2} - \frac{GML^2}{r^3}, \quad (8.61)$$

As with massive particles, this potential vanishes at the Schwarzschild radius $r = R_s$. For massless particles, except for $L = 0$, the potential exhibits a barrier that particles must overcome to reach the central object. The potential reaches its maximum at $r = 3GM$, corresponding to an unstable circular orbit for photons. Particles with precisely the right energy can orbit momentarily at this radius, but any perturbation will cause them to either escape to infinity or spiral inward.

8.3.1 Free fall

The preceding analysis primarily focused on qualitative discussions of circular orbits. We now turn our attention to the purely radial motion of particles, specifically under the assumption of zero angular momentum. Consider a massive particle in free fall from infinity, characterized by the energy and effective potential:

$$E = 1, \quad V_{\text{eff}} = \frac{1}{2} - \frac{GM}{r}. \quad (8.62)$$

The corresponding equation of motion is given by

$$\frac{1}{2} \dot{r}^2 - \frac{GM}{r} = 0 \Rightarrow \frac{dr}{d\tau} = \pm \sqrt{\frac{R_s}{r}}, \quad (8.63)$$

where the derivative is taken with respect to the proper time τ , and the plus and minus signs denote outward (+) or inward (−) radial motion, respectively. For our purposes, we examine the case of inward motion and thus select the negative sign. Additionally, the time-component of 4-velocity satisfies

$$\frac{dt}{d\tau} \left(1 - \frac{R_s}{r} \right) = 1 \Rightarrow \frac{dt}{d\tau} = \frac{1}{1 - \frac{R_s}{r}}. \quad (8.64)$$

Consequently, the particle's 4-velocity takes the form

$$u^\mu = \left(\left(1 - \frac{R_s}{r} \right)^{-1}, -\sqrt{\frac{R_s}{r}}, 0, 0 \right). \quad (8.65)$$

Integrating the radial equation of motion yields

$$\begin{aligned} \sqrt{r} dr &= -\sqrt{R_s} d\tau \quad (\text{“-” : inward motion}) \\ \Rightarrow \Delta\tau &= \frac{2}{3\sqrt{R_s}} (r_0^{3/2} - r^{3/2}), \end{aligned} \quad (8.66)$$

where $\Delta\tau$ is the proper time elapsed as the particle travels from the initial radial coordinate r_0 to the final position r . Remarkably, the proper time required for the particle to reach either the Schwarzschild radius $r = R_s$ or the central singularity $r = 0$ remains finite in both cases.

On the other hand, for a static observer at infinity, the observed time corresponds to the coordinate time t . The relationship between t and the radial motion is governed by the differential equation

$$\frac{dt}{dr} = \frac{dt}{d\tau} \bigg/ \frac{dr}{d\tau} = - \left(\frac{R_s}{r} \right)^{-1/2} \left(1 - \frac{R_s}{r} \right)^{-1}. \quad (8.67)$$

After integration, we get

$$\begin{aligned} \Delta t &= \frac{2}{2\sqrt{R_s}} (r_0^{3/2} - r^{3/2}) + 2\sqrt{R_s} (r_0^{1/2} - r^{1/2}) \\ &\quad + R_s \ln \left| \left(\frac{\sqrt{r} + \sqrt{R_s}}{\sqrt{r} - \sqrt{R_s}} \right) \left(\frac{\sqrt{r_0} - \sqrt{R_s}}{\sqrt{r_0} + \sqrt{R_s}} \right) \right|. \end{aligned}$$

It gives the total coordinate time required for the particle to travel from the initial position r_0 to the final position r . As the particle approaches the Schwarzschild radius $r \rightarrow R_s$, the logarithmic term diverges, leading to $\Delta t \rightarrow \infty$. This implies that, from the perspective of a distant observer, the particle never actually reaches the Schwarzschild radius within a finite coordinate time. Remarkably, this conclusion is not exclusive to observers at infinity. Any static observer, i.e., one fixed at a constant radial position $r_o > R_s$, will similarly perceive the infalling particle as requiring an infinite time to reach R_s .

8.3.2 Escape velocity

We now revisit the problem of escape velocity, traditionally studied in Newtonian gravity, within the framework of general relativity. Specifically, we determine the minimum initial velocity required for a particle to overcome a

star's gravitational attraction and reach infinity. For simplicity, we assume the star is non-rotating, so the most efficient escape trajectory is purely radial. To ensure the particle reaches infinity, it must possess a conserved energy density of at least $E = 1$. Consequently the radial component of its 4-velocity is given by $u^r = \sqrt{\frac{R_s}{r}}$, while the remaining components are as derived in Section 8.3.1. The initial velocity of the particle is defined relative to a static launcher positioned on the star's surface at radius R . Assuming the launcher remains at rest, its 4-velocity is

$$u_{\text{obs}}^\mu(R) = \left(\left(1 - \frac{R_s}{R}\right)^{-1/2}, 0, 0, 0 \right). \quad (8.68)$$

The energy of the particle, as measured by the launcher, is given by $-\hat{P} \cdot \hat{u}_{\text{obs}}$, leading to

$$\mathcal{E}_o = -mg_{tt}u^t u_{\text{obs}}^t = m \left(1 - \frac{R_s}{R}\right)^{-1/2}. \quad (8.69)$$

In the reference frame of the launcher, this energy corresponds to the relativistic expression for a particle with velocity v , $\mathcal{E}_o = m/\sqrt{1-v^2}$, so the escape velocity of the particle is

$$v_{\text{escape}} = \left(\frac{R_s}{R}\right)^{1/2}. \quad (8.70)$$

This result coincides exactly with the escape velocity derived in Newtonian gravity.

Finally, let us examine the stable circular orbit more carefully. The radius of a stable circular orbit is given by

$$R_+ = \frac{L^2}{R_s} \left(1 + \sqrt{1 - 12 \left(\frac{GM}{L} \right)^2} \right). \quad (8.71)$$

As evident from Eq. (8.71), the smaller the angular momentum, or the smaller L/GM , the smaller R_+ is. The innermost stable circular orbit (ISCO) is at

$$R_{\text{ISCO}} = 6GM. \quad (8.72)$$

For an observer at infinity, the angular velocity of a particle in a circular orbit of radius R_c is

$$\Omega = \frac{d\phi}{dt} = \frac{d\phi/d\tau}{dt/d\tau} = \frac{1}{R_c^2} \left(1 - \frac{R_s}{R_c} \right) \frac{L}{E}. \quad (8.73)$$

Remarkably, it satisfies the relation

$$\Omega^2 = \frac{GM}{R_c^3}, \quad (8.74)$$

which is identical to Kepler's third law for circular orbits in Newtonian mechanics. The particle's 4-velocity takes the form $u^\mu = u^t(1, 0, 0, \Omega)$, where

$$u^t = \left(1 - \frac{R_s}{R_c} - R_c^2 \Omega^2\right)^{-1/2} = \left(1 - \frac{3GM}{R_c}\right)^{-1/2}. \quad (8.75)$$

Consider a stationary observer on the circular orbit, e.g., supported by a powerful rocket to maintain their position. Their 4-velocity is $u_H^\mu = ((1 - R_s/R_c)^{-1/2}, 0, 0, 0)$. The particle has $u_\mu = (-E, 0, 0, L)$, so the energy density γ measured by the observer is

$$\gamma = E_H = -u_\mu u_H^\mu = \frac{E}{(1 - R_s/R_c)^{1/2}}. \quad (8.76)$$

In special relativity, $\gamma = (1 - v^2)^{-1/2}$, so the observed velocity of the particle is $v_H = \sqrt{\frac{R_s/(2R_c)}{1 - R_s/R_c}}$. At $R_c = 2R_s$, this velocity coincides with the radial escape velocity derived earlier. For $R_c < 2R_s$, it exceeds the radial escape velocity, rendering the orbit unstable under perturbations. In such cases, the particle may become unbound and escape to infinity. This conclusion is consistent with our earlier analysis of orbital stability.

8.4 Classical Experimental Tests of General Relativity

In this section, we present a historical overview of key experimental tests validating general relativity. These classical experiments fundamentally rely on particle dynamics in Schwarzschild spacetime. We begin our discussion with the anomalous perihelion precession of Mercury's orbit.

8.4.1 Precession of Mercury's perihelion

Before the advent of general relativity, long-term astronomical observations had revealed an anomalous precession in Mercury's perihelion. Even after accounting for perturbations due to the Sun's non-spherical gravitational potential and the gravitational influences of other planets, a small but persistent discrepancy remained—one that could not be explained within the framework of Newtonian mechanics. At the time, most physicists dismissed this anomaly as insignificant, attributing it to the hypothetical presence of

an undiscovered intra-Mercurial planet. Einstein himself was aware of Mercury's anomalous precession prior to formulating general relativity, though he did not consider it a primary motivation for his theory. However, in late 1915, shortly after completing the field equations of general relativity, Einstein revisited the problem. Remarkably, his theory naturally accounted for the residual precession, resolving a decades-old puzzle in celestial mechanics and providing one of the earliest empirical validations of general relativity.

In Newtonian gravity, a particle moving in a spherically symmetric gravitational potential experiences a central force given by: $\mathbf{F} = -m \frac{GM}{r^3} \mathbf{r}$ leading to the equation of motion $\ddot{\mathbf{r}} = -\frac{GM}{r^3} \mathbf{r}$. Under such a force, the particle's angular momentum $\mathbf{A} = \mathbf{r} \times m\dot{\mathbf{r}}$ is conserved. Defining $\mathbf{A} = m\mathbf{L}$, where $\mathbf{L}$ is a constant vector in both magnitude and direction, we have

$$\mathbf{r} \perp \mathbf{L}. \quad (8.77)$$

This orthogonality condition implies that the particle's motion is confined to a fixed orbital plane, allowing the trajectory to be described in polar coordinates (r, ϕ) . The dynamics are governed by the following equations

$$\begin{aligned} r^2 \dot{\phi} &= L, \\ \ddot{r} - r \dot{\phi}^2 &= -\frac{GM}{r^2}, \end{aligned} \quad (8.78)$$

where $L = |\mathbf{L}|$ is the conserved angular momentum per unit mass. The first equation expresses the conservation of angular momentum, a direct consequence of the central force; while the second equation is derived from Newton's second law in the radial direction. From these two equations, we seek to determine the particle's orbit $r(\phi)$. To simplify the analysis, we introduce the substitution $u = r^{-1}$. Differentiating $r = u^{-1}$ with respect to time and using angular momentum conservation, we have

$$\begin{aligned} \dot{r} &\equiv \frac{dr}{dt} = \frac{d}{dt} \left(\frac{1}{u} \right) = -\frac{1}{u^2} \frac{du}{d\phi} \dot{\phi} \\ &= -\frac{1}{u^2} L u^2 \frac{du}{d\phi} = -L \frac{du}{d\phi}. \end{aligned} \quad (8.79)$$

Taking the time derivative of $\dot{r}$ yields $\ddot{r} = -L^2 u^2 \frac{d^2 u}{d\phi^2}$. Thus, the orbit can

be determined as:

$$\begin{aligned}\frac{d^2u}{d\phi^2} + u &= \frac{GM}{L^2} \quad (\text{Binet's equation}) \\ \Rightarrow u &= \frac{GM}{L^2} + C \cos(\phi - \phi_0) \quad (C, \phi_0 \text{ are constants}) \\ \Rightarrow \frac{l}{r} &= 1 + e \cos(\phi - \phi_0),\end{aligned}\tag{8.80}$$

where

$$l = L^2/GM, \quad e = CL^2/GM.\tag{8.81}$$

This is the equation of a conic section, with l (the semi-latus rectum) determining the size of the orbit, and e known as the eccentricity, determining the shape of the orbit, while ϕ_0 determines the orientation of the orbit (relative to the x -axis). For $0 < e < 1$, the section is an ellipse, confirming that bound planetary trajectories in a spherically symmetric Newtonian potential are closed ellipses. The above relationships can be expressed equivalently as

$$r(\phi) = \frac{(1 - e^2)a}{1 + e \cos(\phi - \phi_0)},\tag{8.82}$$

where a is the semi-major axis. Setting $\phi_0 = 0$ aligns the orbit's pericenter with the x -axis, as shown in Figure 8.3. The closed nature of Newtonian orbits arises from the inverse-square law, which exhibits a hidden symmetry: the conservation of the Laplace–Runge–Lenz (LRL) vector. This symmetry ensures that bound orbits with $e < 1$ are perfectly closed ellipses. If the central force deviates from the inverse-square law, e.g., due to higher-order corrections such as $1/r^3$ term from non-spherical mass distributions or relativistic effects as in general relativity, the LRL vector is no longer conserved, and orbits fail to close. Instead, the pericenter precesses.

In general relativity, even in the spherically symmetric Schwarzschild spacetime, a particle's motion—while capable of forming a circular orbit—does not trace a closed elliptical path as in Newtonian gravity. Instead, the orbit exhibits a characteristic precession, resulting in a rosette-like pattern. The geodesic equation derived earlier is given by

$$\dot{r}^2 + \left(1 - \frac{R_s}{r}\right) \left(\frac{L^2}{r^2} + 1\right) = E^2.\tag{8.83}$$

To analyze the shape of the orbit rather than merely its radial evolution, we utilize the conserved angular momentum density. Using the chain rule,

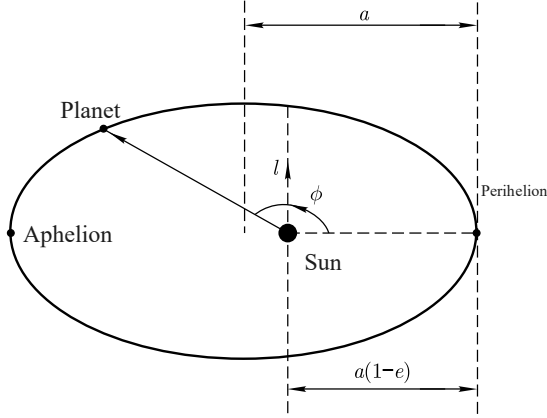


Fig. 8.3 In Newtonian mechanics, if only the two-body interaction between the sun and the planet is considered, then the closed orbit of the planet around the sun will be a strict ellipse. In the figure a is called the semi-major axis of the ellipse, the perihelion distance is $r_- = a(1 - e)$, and the aphelion distance is $r_+ = a(1 + e)$. By determining perihelion and aphelion, we can determine the semi-major axis and eccentricity of the orbit, and thus the entire orbit.

we find $\dot{r} = \frac{dr}{d\lambda} = \frac{dr}{d\phi} \frac{d\phi}{d\lambda} = \frac{dr}{d\phi} \dot{\phi}$, and then obtain the orbital equation

$$\left(\frac{dr}{d\phi}\right)^2 + \left(1 - \frac{R_s}{r}\right) r^2 \left(1 + \frac{r^2}{L^2}\right) = E^2 \frac{r^4}{L^2}. \quad (8.84)$$

Introducing the dimensionless variable $x \equiv \frac{2L^2}{R_s r}$, the equation of motion simplifies to

$$\left(\frac{dx}{d\phi}\right)^2 + \left(\left(\frac{2L}{R_s}\right)^2 - \frac{4E^2 L^2}{R_s^2}\right) - 2x + x^2 - \frac{R_s^2}{2L^2} x^3 = 0. \quad (8.85)$$

Here the last term on the left side of the equation represents a small relativistic perturbation to the Newtonian orbit. Differentiating this first-order equation once more with respect to $\frac{d}{d\phi}$, we have

$$\begin{aligned} & 2 \frac{d}{d\phi} \left(\frac{dx}{d\phi}\right) \frac{dx}{d\phi} - 2 \frac{dx}{d\phi} + 2x \frac{dx}{d\phi} - \frac{3R_s^2}{2L^2} x^2 \frac{dx}{d\phi} = 0 \\ \Rightarrow & \left(2 \frac{d^2 x}{d\phi^2} - 2 + 2x - \frac{3R_s^2}{2L^2} x^2\right) \frac{dx}{d\phi} = 0 \\ \Rightarrow & \frac{d^2 x}{d\phi^2} - 1 + x = \frac{3R_s^2}{4L^2} x^2. \end{aligned} \quad (8.86)$$

In the absence of the right-hand term, the final equation reduces to the normalized Binet equation from Newtonian gravity, which yields conic-section trajectories (e.g., closed ellipses). The inclusion of the right-hand term introduces a general relativistic correction, necessitating a perturbative approach. We decompose the solution as $x = x_0 + x_1$, where

$$\begin{aligned}\frac{d^2 x_0}{d\phi^2} - 1 + x_0 &= 0, \\ \frac{d^2 x_1}{d\phi^2} + x_1 &= \frac{3R_s^2}{4L^2} x_0^2.\end{aligned}\tag{8.87}$$

Here x_0 describes the Newtonian elliptical orbit, while x_1 encodes the relativistic perturbation. As previously established, the zeroth-order solution is $x_0 = 1 + e \cos \phi$. Substituting this into the equation for x_1 , we obtain

$$\frac{d^2 x_1}{d\phi^2} + x_1 = \alpha(1 + e^2 \cos^2 \phi + 2e \cos \phi),$$

where $\alpha = \frac{3R_s^2}{4L^2}$. The exact solution for x_1 is

$$x_1 = \alpha \left(1 + \frac{e^2}{2} - \frac{e^2 \cos 2\phi}{6} + e\phi \sin \phi \right).\tag{8.88}$$

In this solution, $1 + \frac{e^2}{2}$ represents a minor shift in the orbit's center, and the last term $\frac{e^2}{6} \cos 2\phi$ describes a small oscillation with no cumulative effect, only the term proportional to $e\phi \sin \phi$ leads to precession of the orbit. Retaining only the precession-relevant term, the combined solution becomes

$$\begin{aligned}x &= 1 + e \cos \phi + \alpha e \phi \sin \phi \\ &\approx 1 + e \cos(\phi - \alpha \phi),\end{aligned}\tag{8.89}$$

where the approximation holds for $\alpha \ll 1$. The orbit remains approximately elliptical but now exhibits a shifted period. Instead of closing after 2π , the perihelion advances by

$$\Delta\phi = 2\pi\alpha = \frac{6\pi G^2 M^2}{L^2 c^4},\tag{8.90}$$

per revolution. This phenomenon, known as perihelion precession, manifests as a gradual rotation of the orbit's major axis (Figure 8.4). For elliptical orbits, the radial coordinate r is given by

$$r = \frac{2L^2}{R_s} \frac{1}{1 + e \cos \phi} = \frac{(1 - e^2)a}{1 + e \cos \phi},\tag{8.91}$$

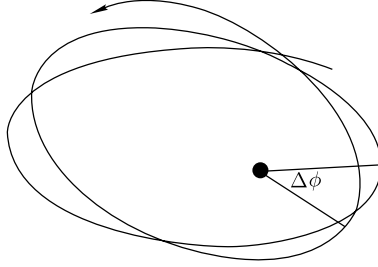


Fig. 8.4 Schematic diagram of planetary perihelion precession.

where

$$a \frac{1}{1-e^2} \frac{2L^2}{R_s} \quad (8.92)$$

is the semi-major axis of the ellipse. The perihelion (closest approach) and aphelion (farthest point) of the orbit are determined by the condition $dr/d\phi = 0$. From the orbital equation, we find that the perihelion occurs at $\phi = 0$:

$$\left. \frac{dr}{d\phi}(r_+) \right|_{\phi=0} = 0 \Rightarrow r_+ = \frac{2L^2}{R_s(1+e)} = a(1-e); \quad (8.93)$$

while the aphelion is at $\phi = \pi$:

$$\left. \frac{dr}{d\phi}(r_-) \right|_{\phi=\pi} = 0 \Rightarrow r_- = \frac{2L^2}{R_s(1-e)} = a(1+e). \quad (8.94)$$

These relations show that the semi-major axis a can be determined observationally from the perihelion and aphelion distances $a = \frac{(r_+ + r_-)}{2}$. Using a and $L^2 = GM(1-e^2)a$, we get the precession angle of the planet's perihelion advance per orbit:

$$\Delta\phi = \frac{6\pi GM}{(1-e^2)a}. \quad (8.95)$$

For the Sun, with $GM_\odot = 1.475 \text{ km}$, Mercury's orbital period is 88 days, and the precession advance per orbit is $\Delta\phi = 0.1038''$. While this per-orbit precession is minuscule, its cumulative effect over a century becomes significant due to Mercury's high orbital frequency $\Delta\phi|_{\text{century}} = 42.98''$. Modern measurements of Mercury's perihelion advance yield $43.1'' \pm 0.5''$ per century, in striking agreement with general relativity's prediction. This precise match (within 0.3% uncertainty) stands as one of the theory's earliest and most compelling experimental validations.

Table 8.1 The perihelion precession of some planets in the solar system (unit: "/century).

Planet	General Relativity Prediction	Experimental Observation
Venus	8.6	8.4 ± 4.8
Earth	3.8	5.0 ± 1.2
Icarus	10.3	9.8 ± 0.8

All bound orbits in the solar system exhibit this relativistic precession when modeled in the Sun's spherically symmetric spacetime. Table 8.1 summarizes theoretical and observed precession rates for select solar system bodies, highlighting the consistency of general relativity across diverse orbital configurations.

As discussed above, the precession of a planet's perihelion in general relativity arises from a correction term to Newtonian gravity. From the perspective of classical mechanics, this relativistic correction can be interpreted as the introduction of a small non-inverse-square force. In fact, various astrophysical effects can produce additional non-inverse-square forces, leading to further contributions to perihelion precession. For instance, if the Sun deviates from a perfect sphere, its gravitational potential acquires multipole moment corrections. In the equatorial plane ($\theta = \pi/2$), this potential takes the form

$$\Phi = -\frac{GM}{r} - \frac{1}{2} \frac{GM J_2 R_\odot^2}{r^3}, \quad (8.96)$$

where J_2 quantifies the quadrupole moment contribution. Comparing this with the general relativistic correction derived earlier, we obtain the quadrupole-induced precession:

$$\begin{aligned} \Delta\phi_{\text{quad}} &= \frac{6\pi GM}{(1-e^2)a} \frac{1}{2} \frac{J_2 R_\odot^2}{GM(1-e^2)a} \\ &= \frac{1}{2} \frac{J_2 R_\odot^2}{GM(1-e^2)a} \Delta\phi_{\text{GR}}. \end{aligned} \quad (8.97)$$

For Mercury, the accumulated contribution to the perihelion precession over a century is

$$\Delta\phi|_{\text{century}} = 42.98''(1 + 3 \times 10^{-4}(J_2/10^{-7})). \quad (8.98)$$

Observational constraints indicate $J_2 \sim 10^{-7}$ for the Sun, rendering this effect negligible compared to the dominant general relativistic precession.

In fact, the gravitational perturbations from other planets exert a far more significant influence on Mercury's orbit than the relativistic or solar quadrupole effects. Through meticulous perturbation calculations in the late 19th century, astronomers derived the contributions to Mercury's perihelion precession summarized in Table 8.2.

Table 8.2 The influence of other planets in the solar system on the precession of Mercury's perihelion (unit: ''/century).

Perturbing Planet	Precession of Mercury's Perihelion
Venus	277.86 ± 0.27
Earth	90.04 ± 0.08
Mars	2.54
Jupiter	153.58
Saturn	7.30
Uranus	0.14
Neptune	0.04
Total	531.5 ± 0.3

8.4.2 Gravitational redshift

In Chapter 3, we demonstrated that photons experience frequency shifts when propagating through a gravitational potential. This effect was derived using the equivalence principle and simple thought experiments, yielding a redshift factor $z = \Delta\phi$ proportional to the gravitational potential difference. In this section, we present a rigorous treatment of gravitational redshift within the framework of curved spacetime, elucidating its geometric origin in general relativity.

Let us first examine a simple scenario for gravitational redshift. Consider a photon emitter fixed at coordinates (r_E, θ_E, ϕ_E) , and a receiver is fixed at (r_R, θ_R, ϕ_R) . Photons can be regarded as waves, where the distance between two successive wave crests (or troughs) determines their wavelength and frequency. The first wave crest travels from the emitter to the receiver along a null geodesic $(t_E, \theta_E, \phi_E) \rightarrow (t_R, \theta_R, \phi_R)$, and the second crest follows a similar trajectory but is emitted at a slightly later time $(t_E + \Delta t_E) \rightarrow (t_R + \Delta t_R)$. In Schwarzschild spacetime, the null condition

$$ds^2 = 0 \quad \Rightarrow \quad \left(1 - \frac{R_s}{r}\right) dt^2 = \left(1 - \frac{R_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

Parameterizing the null curve with an affine parameter λ , we obtain

$$\begin{aligned} \frac{dt}{d\lambda} &= \left(1 - \frac{R_s}{r}\right)^{-1/2} \left(g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}\right)^{1/2} \\ \Rightarrow t_R - t_E &= \int_{\lambda_E}^{\lambda_R} \left(1 - \frac{R_s}{r}\right)^{-1/2} \left(g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}\right)^{1/2} d\lambda. \end{aligned}$$

For spatially fixed emitters and receivers, the coordinate time difference $t_R - t_E$ is identical for all photons. Consequently, the time intervals between successive crests satisfy $\Delta t_E = \Delta t_R$ for an observer at rest at infinity.

However, the proper time intervals measured by the emitter and receiver differ due to gravitational time dilation. Since $dr = d\theta = d\phi = 0$, the proper time is given by $d\tau^2 = \left(1 - \frac{R_s}{r}\right) dt^2$, leading to

$$\Delta\tau_E = \left(1 - \frac{R_s}{r_E}\right)^{1/2} \Delta t_E, \quad \Delta\tau_R = \left(1 - \frac{R_s}{r_R}\right)^{1/2} \Delta t_R.$$

The ratio of these intervals is

$$\frac{\Delta\tau_R}{\Delta\tau_E} = \left(\frac{1 - R_s/r_R}{1 - R_s/r_E}\right)^{1/2}. \quad (8.99)$$

Here $\Delta\tau$ represents the period T between wave crests as measured locally. Thus, the frequency ratio observed by the receiver and emitter is

$$\frac{\nu_R}{\nu_E} = \left(\frac{1 - R_s/r_E}{1 - R_s/r_R}\right)^{1/2}. \quad (8.100)$$

If $r_R > r_E$, then $\nu_R < \nu_E$, the photon is redshifted, and the redshift factor is

$$1 + z = (\nu_R/\nu_E)^{-1} \Rightarrow z = \sqrt{\frac{1 - R_s/r_R}{1 - R_s/r_E}} - 1. \quad (8.101)$$

In the weak field limit, $R_s/r \ll 1$, this expression simplifies to

$$\begin{aligned} z &= \frac{1}{2} \left(\frac{R_s}{r_E} - \frac{R_s}{r_R} \right) = \frac{GM}{r_E} - \frac{GM}{r_R} \\ &= -\Phi(r_E) + \Phi(r_R) = \Delta\Phi, \end{aligned} \quad (8.102)$$

where Φ is the Newtonian gravitational potential. This matches our earlier result derived from the equivalence principle. Crucially, the agreement holds only in the weak-field regime, highlighting the necessity of general relativity for strong gravitational fields.

In fact, for a spacetime with the metric

$$ds^2 = g_{00}(\mathbf{x})dt^2 + g_{ij}(\mathbf{x})dx^i dx^j, \quad (8.103)$$

if the emitter and receiver are fixed, then

$$\frac{\nu_R}{\nu_E} = \left(\frac{g_{00}(\mathbf{x}_E)}{g_{00}(\mathbf{x}_R)} \right)^{1/2}. \quad (8.104)$$

More generally, when the emitter and receiver are not fixed in space but undergo motion, such as in free fall, how to compute the redshift? This scenario can be addressed using the general framework for measurements introduced earlier. Consider a spacetime manifold with the metric $g_{\mu\nu}$,

where the emitter is located at point A and the receiver at point B . The energies of the photons detected by each are given by

$$\begin{aligned}\mathcal{E}_E(A) &= -\hat{P}(A) \cdot \hat{u}_E(A) = -P_\mu(A)u_E^\mu(A), \\ \mathcal{E}_R(B) &= -\hat{P}(B) \cdot \hat{u}_R(B) = -P_\mu(B)u_R^\mu(B),\end{aligned}\tag{8.105}$$

where $\hat{P}$ is the photon's 4-momentum. Consequently, the ratio of the observed photon frequencies is

$$\frac{\nu_R}{\nu_E} = \frac{P_\mu(B)u_R^\mu(B)}{P_\mu(A)u_E^\mu(A)}.\tag{8.106}$$

If $P_\mu(A)$ is known, $P_\mu(B)$ can be determined via the null geodesic equation:

$$\frac{dP_\mu}{d\lambda} - \Gamma_{\mu\rho}^\nu P_\nu \frac{dx^\rho}{d\lambda} = 0 \Rightarrow \frac{dP_\mu}{d\lambda} = \Gamma_{\mu\rho}^\nu P_\nu P^\rho.\tag{8.107}$$

For a stationary spacetime, i.e., one admitting a timelike Killing vector ∂_t in some coordinate system, the time component P_0 of the photon's 4-momentum is conserved. If both the emitter and receiver are spatially fixed, their spatial components of 4-velocity are both zero:

$$u_E^i = 0, \quad u_R^i = 0.\tag{8.108}$$

Using the normalization condition of the 4-velocities of the emitter and receiver $g_{\mu\nu}u^\mu u^\nu = -1$, we get

$$u^0 = \frac{1}{\sqrt{-g_{00}}}.\tag{8.109}$$

Thus, the frequency ratio simplifies to

$$\frac{\nu_R}{\nu_E} = \frac{P_0(B)}{P_0(A)} \left(\frac{g_{00}(A)}{g_{00}(B)} \right)^{1/2} = \left(\frac{g_{00}(A)}{g_{00}(B)} \right)^{1/2},\tag{8.110}$$

recovering our earlier result. More generally, when the emitter or receiver is in motion, the analysis becomes more intricate. We will examine this situation in detail in Section 8.6 within the context of relativistic astrophysics.

8.4.3 Null geodesics

Among the classical tests of general relativity, two key experiments—the deflection of light and the radar echo delay—are intimately connected to the motion of photons in Schwarzschild spacetime. To understand these phenomena, we must first carefully analyze the behavior of null geodesics in this spacetime.

Null geodesics in Schwarzschild spacetime satisfy the equation

$$-\left(1 - \frac{R_s}{r}\right)^{-1} E^2 + \left(1 - \frac{R_s}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} = 0, \quad (8.111)$$

where E and L are the conserved quantities associated with the photon's "energy" and "angular momentum," respectively:

$$E = \left(1 - \frac{R_s}{r}\right) \frac{dt}{d\lambda}, \quad L = r^2 \frac{d\phi}{d\lambda}.$$

Since the affine parameter λ is arbitrary, these quantities lack an unambiguous physical interpretation. To resolve this ambiguity, we introduce the impact parameter b , defined as

$$b^2 = \frac{L^2}{E^2}, \quad (8.112)$$

which eliminates the dependence on λ . Substituting this into (8.111), we obtain

$$\frac{1}{b^2} = \frac{1}{L^2} \left(\frac{dr}{d\lambda}\right)^2 + W_{\text{eff}}(r), \quad (8.113)$$

where the effective potential is given by:

$$W_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{R_s}{r}\right). \quad (8.114)$$

Equation (8.113) resembles the energy equation for a particle in a potential, with $1/b^2 \sim E$ playing the role of an effective energy and W_{eff} acting as the potential. Notably, the affine parameter λ be rescaled $\lambda \rightarrow L\lambda$ to absorb L , so the equation only depends solely on b^2 . While E and L are coordinate-dependent and lack direct physical meaning, the impact parameter b is a gauge-invariant quantity with a clear geometric interpretation, it characterizes the photon's trajectory relative to the central mass. We define the impact parameter as $b = |L/E|$. Working in the equatorial plane $\theta = \pi/2$, we employ polar coordinates $x = r \cos \phi$, $y = r \sin \phi$. In the asymptotic limit $r \rightarrow \infty$, we find

$$b = |L/E| = \frac{r^2 d\phi/d\lambda}{dt/d\lambda} = r^2 \frac{d\phi}{dt}. \quad (8.115)$$

At infinity, the photon's trajectory approximates a straight line with $\phi \approx \frac{d}{r}$, where d is the minimal distance between the light ray and the central mass, and $\frac{dr}{dt} \approx -1$ for ingoing photon. This yields

$$\frac{d\phi}{dt} = \frac{d}{r^2} \quad \Rightarrow \quad b = d. \quad (8.116)$$

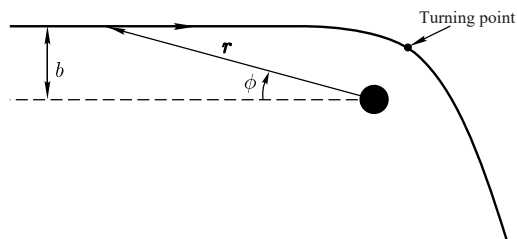


Fig. 8.5 The physical meaning of the impact parameter.

Thus, b indeed corresponds to the impact parameter: with the star's center as the origin, the light ray is approximately a straight line at asymptotic infinity, whose minimum distance from the central object is precisely b , as shown in Figure 8.5. Similar to massive particles, photon motion can be described using an effective potential $W_{\text{eff}}(r)$. This potential peaks at $r = 3GM$,

$$W_{\text{eff}}(r = 3GM) = \frac{1}{27G^2M^2}. \quad (8.117)$$

This maximum determines the critical impact parameter

$$b_c = \sqrt{27}GM. \quad (8.118)$$

The critical case $b = b_c$ corresponds to an unstable circular photon orbit. When $b > b_c$, the photon's "energy" $1/b^2$ lies below the potential barrier, resulting in a scattering trajectory with a turning point. The photon approaches from infinity, is deflected, and escapes back to infinity. When $b < b_c$, the photon overcomes the potential barrier and is captured by the central object. The corresponding capture cross-section is

$$\sigma = \pi b_c^2 = 27\pi(GM)^2. \quad (8.119)$$

For distant observers, the black hole's effective capture radius appears as $3\sqrt{3}GM = \frac{3\sqrt{3}}{2}R_s$. Any photon with impact parameter $b < 3\sqrt{3}GM$ will inevitably be captured.

We can also examine the outward photon orbit outside the event horizon, within the range $R_s < r < 3GM$. As we will later demonstrate, photons inside the horizon $r \leq R_s$ cannot escape due to the causal structure of spacetime. In the range $R_s < r < 3GM$, photons must overcome a potential barrier to propagate to infinity. This scenario closely parallels our earlier discussion. For $b^2 < 27G^2M^2$, photons possess sufficient energy

to surmount the barrier and escape to infinity. These photons have relatively small angular momentum, favoring radial motion. Conversely, for $b^2 > 27G^2M^2$, photons fail to overcome the barrier and ultimately fall back into the black hole after reaching a turning point. Here, the larger angular momentum prevents escape from the gravitational pull.

An interesting question arises. Suppose a transmitter is positioned at radial position $r = R$ in the plane $\theta = \pi/2$, emitting photons at an angle ψ relative to the radial direction. If $\psi = 0$, $b = 0$, the photons strictly move radially, and they will always escape to infinity. The question is, what is the critical emission angle ψ_c beyond which photons can no longer escape? In the transmitter's local reference frame, $\hat{e}_0$ is its timelike 4-velocity, and the other three spacelike basis vectors are mutually orthogonal and normalized,

$$\hat{e}_r = (0, (1 - R_s/R)^{1/2}, 0, 0), \quad \hat{e}_0 = (0, 0, 0, 1/R). \quad (8.120)$$

The emission angle relative to the transmitter satisfies

$$\tan \psi = \frac{\hat{u} \cdot \hat{e}_\phi}{\hat{u} \cdot \hat{e}_r}, \quad (8.121)$$

where $\hat{u}$ is the photon's 4-momentum, with components

$$\begin{aligned} u^t &= \frac{dt}{d\lambda} = \frac{E}{1 - R_s/r}, \\ u^r &= \frac{dr}{d\lambda} = L \left(\frac{1}{b^2} - W_{\text{eff}}(r) \right)^{1/2}, \\ u^\theta &= \frac{d\theta}{d\lambda} = 0, \\ u^\phi &= \frac{d\phi}{d\lambda} = \frac{L}{r^2}. \end{aligned} \quad (8.122)$$

Since $\hat{u} \cdot \hat{e}_\phi = g_{\phi\phi}(\hat{e}_\phi)^\phi u^\phi = \frac{L}{R}$, and

$$\hat{u} \cdot \hat{e}_r = g_{rr}(\hat{e}_r)^r u^r = (1 - R_s/R)^{-1/2} L \left(\frac{1}{b^2} - \frac{1}{R^2} \left(1 - \frac{R_s}{R} \right) \right)^{1/2}, \quad (8.123)$$

the critical angle ψ_c for $b^2 = 27G^2M^2$ is

$$\tan \psi_c = \frac{1}{R} (1 - R_s/R)^{1/2} \left(\frac{1}{27G^2M^2} - \frac{1}{R^2} \left(1 - \frac{R_s}{R} \right) \right)^{-1/2}.$$

When the transmitter is at $R = 3GM$, $\psi_c = \frac{\pi}{2}$, which corresponds to the unstable circular orbit. A slight outward perturbation allows photon escape. When the transmitter is at $R < 3GM$, ψ_c decreases, narrowing the

escape cone. And when $R = R_s$, $\psi_c = 0$, meaning no escape is possible. Notably, ψ_c . Note that the angle here is actually a conical solid angle. defines a conical solid angle: any photon emitted within this angle (relative to the radial direction) can escape to infinity, regardless of its azimuthal orientation.

A closely related question is: for a stationary observer at a fixed radius $r = R$, maintained by a powerful rocket, what portion of the sky remains visible? This scenario mirrors our earlier discussion of outward photon orbits, but now we must consider inward-propagating null geodesics. The critical angle for visibility is analogous, with the key distinction being the reversed propagation direction. The critical angle satisfies

$$\sin \psi_c = (1 - R_s/R)^{1/2} \frac{b_c}{R}. \quad (8.124)$$

When the angular momentum of the light ray is too large, it cannot be seen by the observer. As the observer approaches the event horizon $R \rightarrow R_s$, the visible sky contracts to a narrow circular disk of angular radius

$$\sin \psi_c \approx (1 - R_s/R)^{1/2} \frac{\sqrt{27}}{2}. \quad (8.125)$$

The disk shrinks as $R \rightarrow R_s$, eventually vanishing entirely. On the other hand, infalling photons are increasingly blueshifted, causing the remaining visible portion of the sky to appear intensely bright.

8.4.4 Light deflection and radar echo delay

The phenomenon of light deflection, first predicted by Einstein's general theory of relativity, was famously confirmed during the 1919 total solar eclipse observation. This gravitational lensing effect becomes particularly significant in strong gravitational fields, making it a powerful tool in modern astrophysics. As previously discussed, when a photon's impact parameter b exceeds the critical value b_c , its trajectory features a turning point at R_0 , where $dr/d\phi = 0$. The photon's path can then be described as originating from infinity, reaching the turning point, and returning to infinity. The orbital equation at this turning point is given by:

$$\frac{d\phi}{dr} = \frac{1}{r^2} \left(\frac{1}{b^2} - \frac{1}{r^3} (r - R_s) \right)^{-1/2}. \quad (8.126)$$

The turning point radius R_0 satisfies the cubic equation

$$R_0^3 - b^2(R_0 - R_s) = 0 \quad \Rightarrow \quad R_0 = \frac{2b}{\sqrt{3}} \cos \left(\frac{1}{3} \cos^{-1} \left(-\frac{b_c}{b} \right) \right). \quad (8.127)$$

The total deflection angle can be obtained by integrating Eq. (8.126) over the photon's trajectory. By symmetry, the deflection angle $\Delta\phi$ receives equal contributions from both the incoming and outgoing paths, yielding

$$\Delta\phi = 2 \int_{R_0}^{\infty} \frac{dr}{(r^4 b^{-2} - r(r - R_s))^{1/2}}. \quad (8.128)$$

In the absence of mass, $M = 0$, we recover the expected straight-line solution

$$\Delta\phi|_{M=0} = 2 \sin^{-1}(b/R_0)|_{R_0=b} = \pi. \quad (8.129)$$

For $M \neq 0$ and $\Delta\phi \neq \pi$, this indicates that due to the gravitational attraction in the Schwarzschild spacetime, the light ray is deflected. Transforming to the variable $u = 1/r$, the deflection angle can be written as

$$\Delta\phi = 2 \int_0^{1/R_0} \frac{du}{(b^{-2} - u^2 + R_s u^3)^{1/2}}. \quad (8.130)$$

The integral's complexity arises because for fixed impact parameter b , the upper limit of the integral R_0 depends on M through R_s . We address this by treating M and R_0 as independent variables, comparing deflection angles at constant R_0 rather than constant b . To the first order in M , keeping b or R_0 constant makes no difference, but it will be different at higher orders.

Using the relation $b = \sqrt{\frac{R_0^3}{R_0 - R_s}}$, we express the deflection angle as

$$\Delta\phi(M) = 2 \int_0^{1/R_0} \frac{du}{(R_0^{-2} - R_s R_0^{-3} - u^2 + R_s u^3)^{1/2}}. \quad (8.131)$$

Thus, we can perform a Taylor expansion with respect to M about $M = 0$:

$$\Delta\phi(M) = \Delta\phi(M=0) + \left. \frac{\partial(\Delta\phi)}{\partial M} \right|_{M=0} M + \dots \quad (8.132)$$

The first-order term evaluates to

$$\begin{aligned} \left. \frac{\partial(\Delta\phi)}{\partial M} \right|_{M=0} &= 2 \int_0^{1/R_0} \frac{G(R_0^{-3} - u^3) du}{(R_0^{-2} - R_s R_0^{-3} - u^2 + R_s u^3)^{1/2}} \Big|_{M=0} \\ &= 2G \int_0^{1/b} \frac{b^{-3} - u^3}{(b^{-2} - u^2)^{3/2}} du \\ &= 4Gb^{-1}. \end{aligned} \quad (8.133)$$

Thus, the leading-order deflection is

$$\begin{aligned} \delta\phi &= \Delta\phi(M) - \Delta\phi(M=0) \approx M \left. \frac{\partial(\Delta\phi)}{\partial M} \right|_{M=0} = \frac{4GM}{b} \\ &= \frac{4GM}{bc^2}. \end{aligned} \quad (8.134)$$

The deflection angle is inversely proportional to b , the closer to the star, the greater the influence of the star, and thus the larger the deflection angle. On the other hand, it is directly proportional to M , the heavier the star, the more pronounced the light deflection. It is also directly proportional to the Newtonian gravitational constant, the larger the Newtonian gravitational coupling constant, the stronger the gravity, and therefore the larger the deflection angle.

We now present an alternative derivation of the light deflection angle, employing a perturbative method analogous to that used in calculating the perihelion precession of planetary orbits. Beginning with the null geodesic equation for light in Schwarzschild spacetime, we have:

$$\frac{d^2 x}{d\phi^2} + x = \alpha x^2, \quad (8.135)$$

where $\alpha = \frac{3R_s^2}{4L^2}$ is a small parameter. In Newtonian gravity, the term on the right-hand side does not exist, the solution describes a straight-line trajectory in polar coordinates

$$x = \frac{1}{D} \sin(\phi - \phi_0). \quad (8.136)$$

or equivalently, in terms of the radial coordinate r and impact parameter b ,

$$r = \frac{b}{\sin(\phi - \phi_0)}. \quad (8.137)$$

Here, we adopt the substitution $x = \frac{2L^2}{R_s r}$ and define $D = \frac{R_s}{2EL}$. For small deflections, we treat the right-hand side of (8.135) as a perturbation. Expanding x as $x = x_0 + \alpha x_1 + \mathcal{O}(\alpha^2)$, where $x_0 = \frac{\sin \phi}{D}$ (setting $\phi_0 = 0$ for simplicity), the first-order correction x_1 satisfies

$$x_1'' + x_1 = x_0^2 = \frac{\sin^2 \phi}{D^2}. \quad (8.138)$$

The inhomogeneous solution to this equation is

$$x_1 = \left(1 + C \cos \phi + \frac{1}{3} \cos 2\phi \right) / 2D^2, \quad (8.139)$$

where C is an integration constant determined by boundary conditions. The smallness of the perturbation is ensured by $\alpha/D = 3R_s/2b \ll 1$, consistent

with weak-field observations. Finally, the full perturbative solution to first order is

$$x \approx \frac{\sin \phi}{D} + \frac{\alpha(1 + C \cos \phi + \frac{1}{3} \cos 2\phi)}{2D^2}. \quad (8.140)$$

To compute the deflection angle, we examine the asymptotic behavior as $r \rightarrow \infty$, i.e., $x \rightarrow 0$. The photon's incoming and outgoing directions correspond to the roots of $x(\phi) = 0$ at

$$\phi = -\epsilon_1, \quad \phi = \pi + \epsilon_2, \quad (8.141)$$

where $\epsilon_1, \epsilon_2 \ll 1$. Substituting these into (8.140) and linearizing yields

$$\begin{aligned} -\frac{\epsilon_1}{D} + \frac{\alpha}{2D^2} \left(\frac{4}{3} + C \right) &= 0, \\ -\frac{\epsilon_2}{D} + \frac{\alpha}{2D^2} \left(\frac{4}{3} - C \right) &= 0. \end{aligned} \quad (8.142)$$

The total deflection angle is then

$$\delta\phi = \epsilon_1 + \epsilon_2 = \frac{4\alpha}{3D} = \frac{4GM}{b}, \quad (8.143)$$

recovering the classic result for small-angle light bending. This perturbative treatment is valid only for small deflections. The above discussion only applies when the light deflection is small, and the approximate conditions are valid. For larger deflections (e.g., near photon spheres or ultracompact objects), a non-perturbative analysis of (8.135) is required, typically involving elliptic integrals or numerical methods. Such cases are critical in strong-field gravitational lensing studies, where higher-order terms in α become significant.

For a light ray grazing the solar limb, the theoretical deflection angle is obtained by substituting the solar mass $M_\odot$ and radius $R_\odot$ into (8.143), $\delta\phi \approx 1.75''$. This prediction was first put to the test during the total solar eclipse of May 29, 1919, when two expeditions led by Arthur Eddington and Frank Dyson observed star positions near the obscured Sun. The teams, stationed in Sobral, Brazil, and Príncipe Island (Gulf of Guinea), measured the apparent shifts of starlight due to the Sun's gravitational field. Their results were:

$$\begin{aligned} \delta\phi &= 1.98'' \pm 0.12'' \text{ (Sobral)}, \\ \delta\phi &= 1.60'' \pm 0.3'' \text{ (Príncipe)}. \end{aligned} \quad (8.144)$$

Eddington subsequently declared these observations a confirmation of Einstein's general relativity, marking a pivotal moment in the acceptance of the theory.⁴

The advent of radio astronomy in the late 20th century revolutionized tests of light deflection, eliminating the need for total solar eclipses. For example: every year on October 8th, the quasar 3C279 passes behind the Sun as viewed from Earth. By comparing its radio wave deflection with that of the nearby quasar 3C273 (separated by $> 10^\circ$), astronomers could isolate the gravitational lensing effect. Similar observations of other quasars confirmed the predicted deflection to within 1% accuracy, far surpassing eclipse-based methods. Furthermore, using the Very Long Baseline Interferometer (VLBI) networks to measure the deflection of electromagnetic waves from radio quasars as they pass by the sun, changes as small as $10^{-4}''$ can be detected. The measured value of $\delta\phi$ agrees with general relativity at the level of 0.99997 ± 0.00016 .

We now examine the radar echo delay experiment, first proposed by Irwin Shapiro in 1964. This experiment demonstrates the gravitational time delay effect predicted by general relativity. The experimental configuration consists of a radar signal emitted from Earth at Schwarzschild coordinate R_E , which passes near the Sun to reach a target planet at radius R_P , and then the radar signal returns to the Earth along the same path, as shown in Figure 8.6. The Sun's gravitational field causes a measurable delay in the round-trip travel time of the signal, known as the Shapiro time delay. In Schwarzschild coordinates, the time component of the photon geodesic equation yields

$$\frac{dt}{dr} = \left(1 - \frac{R_s}{r}\right)^{-1} \left(1 - \left(1 - \frac{R_s}{r}\right) \frac{b^2}{r^2}\right)^{-1/2}. \quad (8.145)$$

The total coordinate time (measured by an observer at infinity) for a one-way trip is given by

$$\frac{1}{2} \Delta t = \left(\int_{R_E}^{R_0} dr + \int_{R_0}^{R_P} dr \right) \left(1 - \frac{R_s}{r}\right)^{-1} \left(1 - \left(1 - \frac{R_s}{r}\right) \frac{b^2}{r^2}\right)^{-\frac{1}{2}}.$$

⁴However, modern historians of science doubt that the technical conditions at the time could have achieved this accuracy, especially since the number of photographs taken at the time was limited (only 2 usable images from Príncipe and 7 from Sobral). It was later found that the light deflection experiments based on total solar eclipses yield less consistent results.

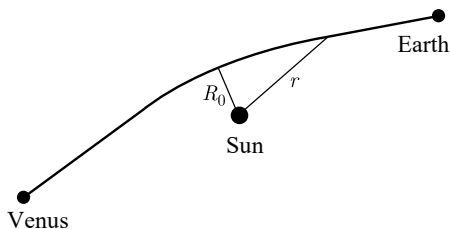


Fig. 8.6 Schematic diagram of radar echo experiment.

If there were no sun, $M = 0$ and $b = R_0$, the time taken would be

$$\begin{aligned}
 \frac{1}{2}\Delta t_0 &= \left(\int_{R_E}^{R_0} dr + \int_{R_0}^{R_P} dr \right) \left(1 - \frac{R_0^2}{r^2} \right)^{-1/2} \\
 &= \left(\int_{R_E}^{R_0} + \int_{R_0}^{R_P} \right) \frac{r dr}{\sqrt{r^2 - R_0^2}} \\
 &= \sqrt{R_E^2 - R_0^2} + \sqrt{R_P^2 - R_0^2}.
 \end{aligned} \tag{8.146}$$

The delayed time is

$$\delta t = \Delta t(M) - \Delta t_0 \approx M \left(\frac{\partial \Delta t}{\partial M} \right) \Big|_{M=0}. \tag{8.147}$$

To first order in M , the full round-trip delay is

$$\begin{aligned}
 \Delta t &= 2 \left(\sqrt{R_E^2 - R_0^2} + \sqrt{R_P^2 - R_0^2} \right) + R_s \left\{ 2 \ln \left(\frac{R_E + \sqrt{R_E^2 - R_0^2}}{R_0} \right) \right. \\
 &\quad \left. + 2 \ln \left(\frac{R_P + \sqrt{R_P^2 - R_0^2}}{R_0} \right) + \left(\frac{R_E - R_0}{R_E + R_0} \right)^{1/2} + \left(\frac{R_P - R_0}{R_P + R_0} \right)^{1/2} \right\}.
 \end{aligned}$$

The first two terms on the right-hand side come from Δt_0 . Since $R_E, R_P \gg R_0$, we obtain

$$\delta t = \frac{4GM}{c^3} \left(\ln \left(\frac{2R_E}{R_0} \right) + \ln \left(\frac{2R_P}{R_0} \right) + 1 \right). \tag{8.148}$$

Note that in the above discussion, we need to vary M and compare geodesics with the same R_0 rather than b . Unlike the case of light deflection, even at the first order in M , whether to keep b or R_0 constant when M changes will affect the final result. The above discussion is for coordinate time. The observed delay on Earth accounts for gravitational time dilation

$$\delta \tau = (1 - R_s/R_E)^{1/2} \delta t. \tag{8.149}$$

In fact, it is difficult to accurately determine R_0 , R_E , and R_P . What we can do is to obtain a formula for how $\Delta\tau$ varies with time (since the Earth and the planet are constantly moving) and treat all parameters as unknowns, then fit the data to determine these parameters.

There are two methodologies employed to measure this effect. One is “passive” observations: waiting for the right opportunity to use other planets, the sun, and the Earth to carry out radar echo experiments. The planet usually chosen is Venus. The theoretical prediction is $\delta\tau \approx 220 \mu\text{s}$. Shapiro’s experiments in 1968 and 1971 confirmed the effect within 2% accuracy. The other is “active” experiments: using spacecraft as precise reflectors to carry out radar echo experiments. It was Anderson who first carried out experiments using spacecraft in 1975. Shapiro’s experiments using the Viking spacecraft in 1977 and 1979 obtained results that deviate from the theoretical prediction by only about 0.1%. In 2002, experiments using the Cassini spacecraft further reduced the deviation between theory and experiment to one in a hundred thousand.

8.4.5 Gravitational time dilation effect

The preceding discussion introduced the classical experimental tests of general relativity. However, numerous additional experiments have been designed to further verify the theory’s predictions. In this section, we focus on the gravitational time dilation effect, which arises from differences in proper time measured by clocks carried by observers in different states of motion. Consider two initially synchronized clocks at the equator: one placed aboard a spacecraft orbiting Earth, and the other remaining stationary on the ground. Assuming $\theta = \pi/2$ and constant r , the spacetime interval is given by

$$ds^2 = - \left(1 - \frac{2GM}{rc^2} \right) c^2 dt^2 + r^2 d\phi^2 = -c^2 d\tau^2. \quad (8.150)$$

For observer A on the ground, moving with Earth’s angular velocity⁵ $\omega = \frac{d\phi}{dt}$, the proper time is

$$d\tau_A^2 = \left(1 - \frac{2GM}{Rc^2} - \frac{R^2\omega^2}{c^2} \right) dt^2, \quad (8.151)$$

⁵In this section we do not consider the effect of the rotation of the Earth on spacetime, assuming the exterior geometry is described by the Schwarzschild metric; we will later account for frame-dragging effects.

where R is Earth's radius. Given the smallness of the dimensionless quantities

$$\frac{2GM}{Rc^2} \ll 1, \quad \frac{R^2\omega^2}{c^2} \ll 1, \quad (8.152)$$

we obtain to first order

$$\delta\tau_A = \left(1 - \frac{GM}{Rc^2} - \frac{R^2\omega^2}{2c^2}\right) \delta t. \quad (8.153)$$

For clock B aboard the spacecraft at altitude $h \ll R$, moving with velocity v relative to the ground, its velocity relative to the asymptotic inertial frame is approximately $v + (R + h)\omega$. Thus,

$$d\tau_B^2 = \left(1 - \frac{2GM}{(R + h)c^2} - \frac{(v + (R + h)\omega)^2}{c^2}\right) dt^2. \quad (8.154)$$

Expanding to first order in h/R and neglecting $h\omega$ compared to v , we find

$$\delta\tau_B = \left(1 - \frac{GM}{(R + h)c^2} - \frac{v^2 + R^2\omega^2 + 2R\omega v}{2c^2}\right) \delta t. \quad (8.155)$$

An observer at infinity would perceive clocks A and B separating at some initial moment and reuniting after a coordinate time interval δt . The proper time difference between the clocks is

$$\delta\tau_A - \delta\tau_B = \left(-\frac{GMh}{R^2c^2} + \frac{(2R\omega + v)v}{2c^2}\right) \delta t. \quad (8.156)$$

Defining the fractional time difference as

$$\Delta \equiv \frac{\delta\tau_A - \delta\tau_B}{\delta\tau_A}, \quad (8.157)$$

we find for a spacecraft at $h = 10^4$ m with $v = 300$ m·s⁻¹: If the spacecraft moves eastward, then $\Delta_E = 1.0 \times 10^{-12}$; If the spacecraft moves westward, then $\Delta_W = -2.1 \times 10^{-12}$. Modern cesium atomic clocks achieve precisions of 10^{-13} , making this effect measurable. Indeed, the 1970 Hafele–Keating experiment confirmed gravitational time dilation, with results deviating from theoretical predictions by approximately 10%.

8.4.6 Parametrized post-Newtonian (PPN) limit

From an experimental standpoint, potential modifications to general relativity can be investigated through well-studied solar system tests. For metric-based gravitational theories, the weak-field and slow-motion conditions of the solar system, where $|\Phi|/c^2 \ll 1$ and $v^2/c^2 \ll 1$, allow different theories to be compared within a unified framework. As established,

general relativity predicts that a spherically symmetric vacuum solution is described by the Schwarzschild metric. However, if we relax the assumption of a specific theory and instead consider a general static, spherically symmetric spacetime, the metric takes the form

$$ds^2 = -B(r)dt^2 + A(r)dr^2 + r^2(\sin^2\theta d\phi^2 + d\theta^2). \quad (8.158)$$

In the solar system, we introduce the dimensionless quantity

$$x \equiv \frac{GM_\odot}{rc^2}, \quad (8.159)$$

which characterizes the strength of the gravitational field. The metric functions $B(r)$ and $A(r)$ can then be expanded in powers of x as

$$\begin{aligned} B(r) &= 1 - 2\alpha x + 2(\beta - \gamma)x^2 + \cdots, \\ A(r) &= 1 + 2\gamma x + \cdots, \end{aligned} \quad (8.160)$$

where α , β , and γ are the post-Newtonian parameters. From the previous discussion of the Newtonian limit, we know that the (00)-component of the metric field is related to the Newtonian gravitational potential, while corrections to other components are negligible at leading order. In the weak field approximation of general relativity, these parameters take the values $\alpha = 1$, $\gamma = 1$, $\beta = 1$. Deviations from these values would indicate corrections to or extensions of general relativity. This parametrization scheme is known as the Parametrized Post-Newtonian (PPN) formalism, which provides a model-independent framework for testing gravitational theories. The expansion can, in principle, be extended to higher orders, yielding post-post-Newtonian corrections. Alternatively, the metric can be expressed in isotropic coordinates as⁶

$$\begin{aligned} ds^2 &= - \left(1 - 2\alpha \frac{GM_\odot}{\rho c^2} + 2\beta \left(\frac{GM_\odot}{\rho c^2} \right)^2 + \cdots \right) dt^2 \\ &\quad + \left(1 + 2\gamma \frac{GM_\odot}{\rho c^2} \right) (d\rho^2 + \rho^2 d\Omega^2). \end{aligned} \quad (8.161)$$

The parametrized post-Newtonian (PPN) metric provides a powerful framework for testing and distinguishing between alternative theories of gravity. For spherically symmetric spacetimes, this approach offers a standardized parametrization that enables systematic comparisons with experimental observations. To investigate potential corrections to general

⁶This expansion was first proposed by Sir Arthur S. Eddington and Howard P. Robertson.

relativity, we may fix $\alpha = 1$, as required by the Newtonian limit, while allowing β and γ to deviate from their general relativistic values. Under the assumption that test particles follow geodesic motion, we can recompute key observational effects within this generalized framework. For the precession of the perihelion of planets, in the above spacetime, the precession advance is given by

$$\delta\phi_{\text{prec}} = \frac{1}{3}(2 + 2\gamma - \beta) \frac{6\pi GM}{c^2 a(1 - e^2)}. \quad (8.162)$$

For light deflection, if the impact parameter is still b , the deflection angle becomes

$$\delta\phi_{\text{def}} = \frac{1 + \gamma}{2} \frac{4GM}{c^2 b}. \quad (8.163)$$

Finally, for Shapiro time delay, there is

$$\Delta t_{\text{delay}} = \frac{1 + \gamma}{2} \frac{4GM}{c^3} \left[\ln \left(\frac{4R_E R_P}{R_0^2} \right) + 1 \right]. \quad (8.164)$$

The Parametrized Post-Newtonian (PPN) formalism allows for stringent tests of general relativity through high-precision solar system experiments.

- Using the deflection of light through the sun can constrain the PPN parameter γ :

$$\gamma = 1.007 \pm 0.009. \quad (8.165)$$

- The time delay of radio signals due to gravitational potential (Shapiro effect) provides even tighter constraints. The Viking spacecraft experiment gave

$$\gamma = 1.000 \pm 0.001. \quad (8.166)$$

The Cassini spacecraft experiment between June 6 and July 7, 2002, gave

$$\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5}, \quad (8.167)$$

representing the most precise measurement to date, confirming general relativity at the 10^{-5} level.

- Using the anomalous advance of Mercury's perihelion constrains the parameter β :

$$\beta = 1.000 \pm 0.003. \quad (8.168)$$

Other experiments based on satellites orbiting the Earth, such as Gravity Probe B, can also give constraints on the PPN parameters.

On the other hand, the PPN limit can serve as an approximation for spherically symmetric spacetime in weak-field regimes. In strongly relativistic systems, such as binary neutron stars or black holes, the weak-field assumption breaks down, but the PPN approach can be generalized to higher orders, e.g. 2PN, 3PN, to model phenomena like periastron advance, spin-orbit coupling, and gravitational wave emission.

8.5 Geodetic Precession

The discussion in the preceding section on experimental tests of general relativity primarily relied on traditional astronomical observations of planetary motion within the solar system. However, advances in space technology have enabled the design of more sophisticated experiments to probe the predictions of general relativity. In this section, we explore the use of gyroscopes to test gravitational effects in the framework of relativity.

In curved spacetime, when a vector is parallel-transported along a closed loop back to its initial position, its final orientation differs from the original due to spacetime curvature. This discrepancy, known as holonomy, provides a measurable manifestation of curvature. Such an effect offers a novel approach to investigating the geometric properties of spacetime. A practical application of this phenomenon involves observing the evolution of the spin vector of a rotating rigid body in a gravitational field.

A spinning probe is referred to as a gyroscope. In the absence of gravitational effects, the spin direction of a gyroscope mounted on a frictionless bracket remains invariant as the bracket moves, meaning the spin undergoes parallel transport. However, since the gyroscope is an object with a scale, the tidal force will cause a tidal torque, which will cause the spin of the gyroscope not to be strictly parallel transported and instead undergo Newtonian (tidal) precession.⁷ The challenge lies in measuring the minuscule relativistic precession predicted by general relativity, as the much larger tidal precession would dominate and obscure the desired effect. To mitigate this, the gyroscope must exhibit exact spherical symmetry,

⁷A well-known example is the equinoxes precession of the Earth's spin, which results from the tidal forces of the sun and moon acting on the bulge of the Earth's equator. If the Earth were strictly spherical, this effect would disappear.